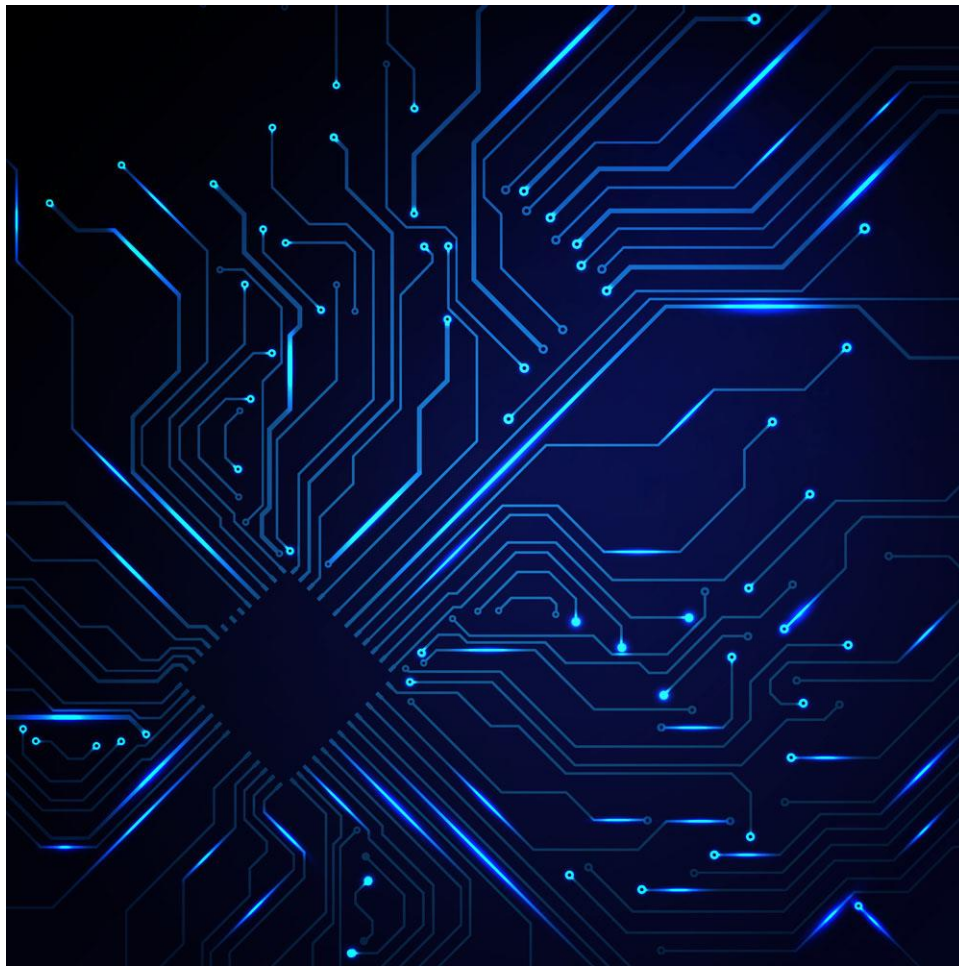




超大型積體電路課程 期末專案

姓名：陳篁瑋

學號：01257024

班級：資工 3A

注意

-
1. 繳交時一律轉 PDF 檔
 2. 繳交期限為
1/7(十七週)
晚上 12 點前
-
3. 一人繳交一份
 4. 檔名請按照作業檔名
格式進行填寫，例
如：
「10857001_陳同學
_HW1」
未依照格式不予批改

※請務必先找助教 demo

※請務必先找助教 demo

※請務必先找助教 demo

需注意的地方：

- 提供的 RISC 沒有使用 SRAM IP
- 01_RTL : testbench.v ('define 部分), CHIP.sv (IOPAD), 01_run.sh(路徑)
- 02_SYN : syn.tcl (讀取 .v 部分)
- 03_GATE_SIM : 01_run_sim.sh(路徑)
- 04_APR : 多個檔案需參考講義實作(CHIP.ioc 等)

以上部分可參考之前作業，自己修改撰寫。

用 Cell-Based 流程進行 IC 設計，並將過程截圖貼上：

(使用 RISC 或自己的程式碼進行)

參考波型(需全部截到)：

1. CHIP module 程式碼截圖：

```

module CHIP(
    input clk,
    input rst,
    output logic[31:0]regs_31
);

logic        ipad_clk;
logic        ipad_reset;
logic [31:0] opad_regs_31;

PDIDGZ_33 u_clk    (.PAD(clk      ),          .C(ipad_clk      )    );
PDIDGZ_33 u_reset  (.PAD(rst      ),          .C(ipad_reset   )    );

PD016CDG_33 u_regs_31_0  (.I(opad_regs_31[0] ), .PAD(regs_31[0] )    );
PD016CDG_33 u_regs_31_1  (.I(opad_regs_31[1] ), .PAD(regs_31[1] )    );
PD016CDG_33 u_regs_31_2  (.I(opad_regs_31[2] ), .PAD(regs_31[2] )    );
PD016CDG_33 u_regs_31_3  (.I(opad_regs_31[3] ), .PAD(regs_31[3] )    );
PD016CDG_33 u_regs_31_4  (.I(opad_regs_31[4] ), .PAD(regs_31[4] )    );
PD016CDG_33 u_regs_31_5  (.I(opad_regs_31[5] ), .PAD(regs_31[5] )    );
PD016CDG_33 u_regs_31_6  (.I(opad_regs_31[6] ), .PAD(regs_31[6] )    );
PD016CDG_33 u_regs_31_7  (.I(opad_regs_31[7] ), .PAD(regs_31[7] )    );
PD016CDG_33 u_regs_31_8  (.I(opad_regs_31[8] ), .PAD(regs_31[8] )    );
PD016CDG_33 u_regs_31_9  (.I(opad_regs_31[9] ), .PAD(regs_31[9] )    );
PD016CDG_33 u_regs_31_10 (.I(opad_regs_31[10] ), .PAD(regs_31[10] )   );
PD016CDG_33 u_regs_31_11 (.I(opad_regs_31[11] ), .PAD(regs_31[11] )   );
PD016CDG_33 u_regs_31_12 (.I(opad_regs_31[12] ), .PAD(regs_31[12] )   );
PD016CDG_33 u_regs_31_13 (.I(opad_regs_31[13] ), .PAD(regs_31[13] )   );
PD016CDG_33 u_regs_31_14 (.I(opad_regs_31[14] ), .PAD(regs_31[14] )   );
PD016CDG_33 u_regs_31_15 (.I(opad_regs_31[15] ), .PAD(regs_31[15] )   );
PD016CDG_33 u_regs_31_16 (.I(opad_regs_31[16] ), .PAD(regs_31[16] )   );
PD016CDG_33 u_regs_31_17 (.I(opad_regs_31[17] ), .PAD(regs_31[17] )   );
PD016CDG_33 u_regs_31_18 (.I(opad_regs_31[18] ), .PAD(regs_31[18] )   );
PD016CDG_33 u_regs_31_19 (.I(opad_regs_31[19] ), .PAD(regs_31[19] )   );
PD016CDG_33 u_regs_31_20 (.I(opad_regs_31[20] ), .PAD(regs_31[20] )   );
PD016CDG_33 u_regs_31_21 (.I(opad_regs_31[21] ), .PAD(regs_31[21] )   );
PD016CDG_33 u_regs_31_22 (.I(opad_regs_31[22] ), .PAD(regs_31[22] )   );
PD016CDG_33 u_regs_31_23 (.I(opad_regs_31[23] ), .PAD(regs_31[23] )   );
PD016CDG_33 u_regs_31_24 (.I(opad_regs_31[24] ), .PAD(regs_31[24] )   );

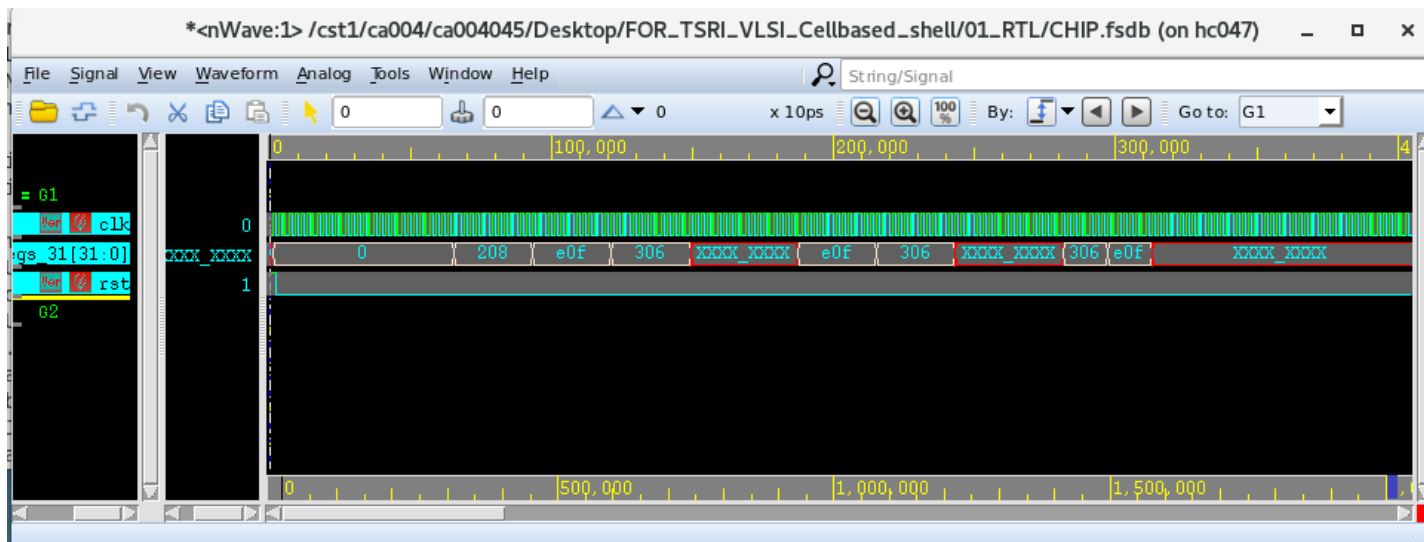
PD016CDG_33 u_regs_31_25 (.I(opad_regs_31[25] ), .PAD(regs_31[25] )   );
PD016CDG_33 u_regs_31_26 (.I(opad_regs_31[26] ), .PAD(regs_31[26] )   );
PD016CDG_33 u_regs_31_27 (.I(opad_regs_31[27] ), .PAD(regs_31[27] )   );
PD016CDG_33 u_regs_31_28 (.I(opad_regs_31[28] ), .PAD(regs_31[28] )   );
PD016CDG_33 u_regs_31_29 (.I(opad_regs_31[29] ), .PAD(regs_31[29] )   );
PD016CDG_33 u_regs_31_30 (.I(opad_regs_31[30] ), .PAD(regs_31[30] )   );
PD016CDG_33 u_regs_31_31 (.I(opad_regs_31[31] ), .PAD(regs_31[31] )   );

RISC_V RISC_V_1(
    .clk      (ipad_clk      ),
    .rst      (ipad_reset   ),
    .regs_31  (opad_regs_31 )
);

endmodule

```

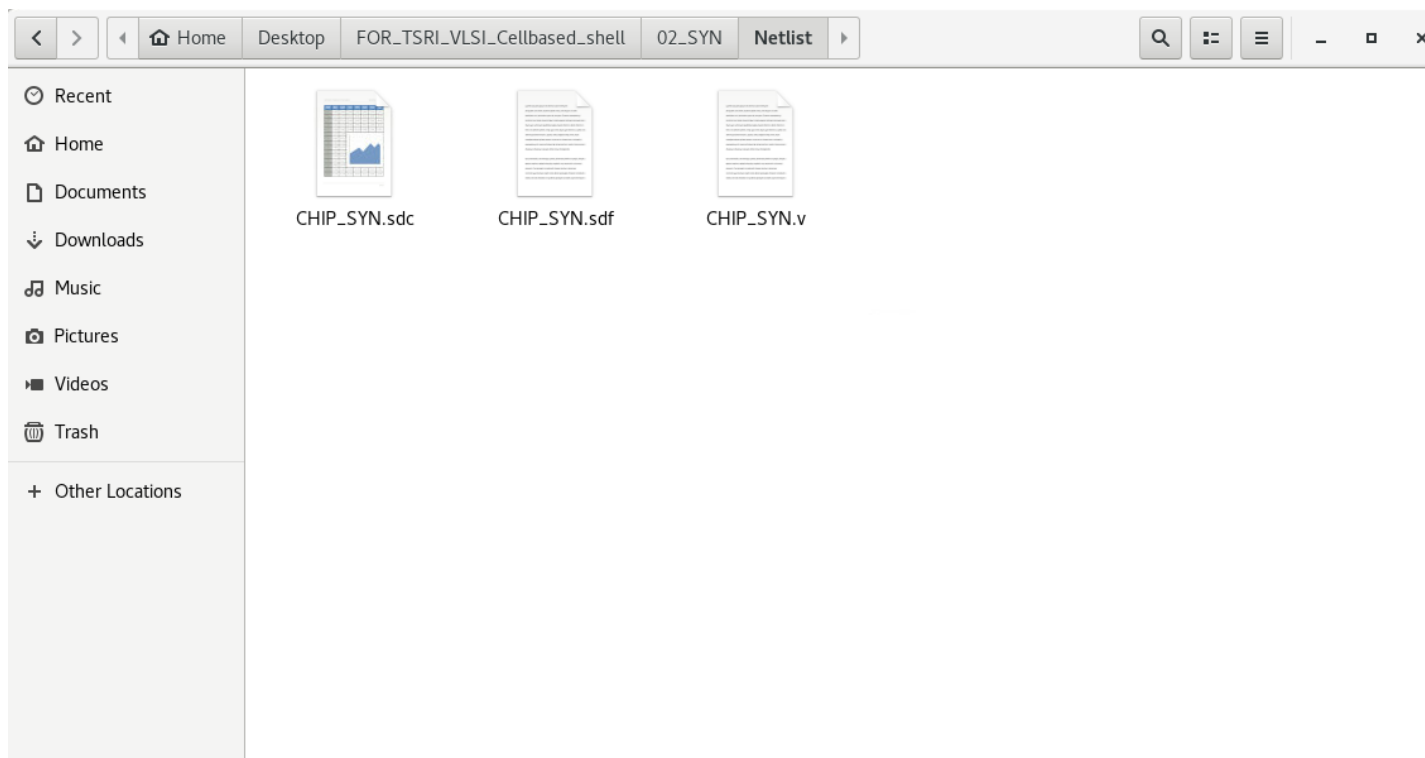
2. RTL 波形模擬結果：



3. Synthesis 的 Timing Report 結果(CHIP.timing 內容)：

CHIP.timing					
~/Desktop/FOR_TSRI_VLSI_Cellbased_shell/02_SYN/Report					
syn.tcl	CHIP.timing		CHIP_SYN.v		
RISC_V_1/LUS_1/ram_0/C2271_17/Z (GTECH_AND2)			0.000	0.000	0.000 r
RISC_V_1/LUS_1/ram_0/N28 (net)	1	0.010	0.000	0.000	0.000 r
RISC_V_1/LUS_1/ram_0/C2271_18/Z (GTECH_AND2)			0.000	0.000	0.000 r
RISC_V_1/LUS_1/ram_0/N29 (net)	1	0.010	0.000	0.000	0.000 r
RISC_V_1/LUS_1/ram_0/C2271_19/Z (GTECH_AND2)			0.000	0.000	0.000 r
RISC_V_1/LUS_1/ram_0/N30 (net)	1	0.010	0.000	0.000	0.000 r
RISC_V_1/LUS_1/ram_0/C2271_20/Z (GTECH_AND2)			0.000	0.000	0.000 r
RISC_V_1/LUS_1/ram_0/N31 (net)	1	0.010	0.000	0.000	0.000 r
RISC_V_1/LUS_1/ram_0/C2271_21/Z (GTECH_AND2)			0.000	0.000	0.000 r
RISC_V_1/LUS_1/ram_0/N32 (net)	1	0.010	0.000	0.000	0.000 r
RISC_V_1/LUS_1/ram_0/C2271_22/Z (GTECH_AND2)			0.000	0.000	0.000 r
RISC_V_1/LUS_1/ram_0/N487 (net)	128	1.280	0.000	0.000	0.000 r
RISC_V_1/LUS_1/ram_0/C2273/Z (GTECH_AND2)			0.000	0.000	0.000 r
RISC_V_1/LUS_1/ram_0/N43 (net)	1	0.010	0.000	0.000	0.000 r
RISC_V_1/LUS_1/ram_0/C2401/Z_127 (*SELECT_OP_2.128_2.1_128)			0.000	0.000	0.000 r
RISC_V_1/LUS_1/ram_0/N298 (net)	8	0.080	0.000	0.000	0.000 r
RISC_V_1/LUS_1/ram_0/ram_reg_0__0_/synch_enable (**SEQGEN**)			0.000	0.000	0.000 r
data arrival time					0.000
clock clk (rise edge)			50.000	50.000	
clock network delay (ideal)			0.000	50.000	
clock uncertainty			-0.100	49.900	
RISC_V_1/LUS_1/ram_0/ram_reg_0__0_/clocked_on (**SEQGEN**)			0.000	49.900	r
library setup time			0.000	49.900	
data required time					49.900
data required time					49.900
data arrival time					0.000
slack (MET)					49.900

4. Synthesis 合成完，Netlist 中的檔案(icon)及.v 檔內容：



```
CHIP_SYN.v
~/Desktop/FOR_TSRI_VLSI_Cellbased_shell/02_SYN/Netlist

syn.tcl x CHIP.timing x CHIP_SYN.v x

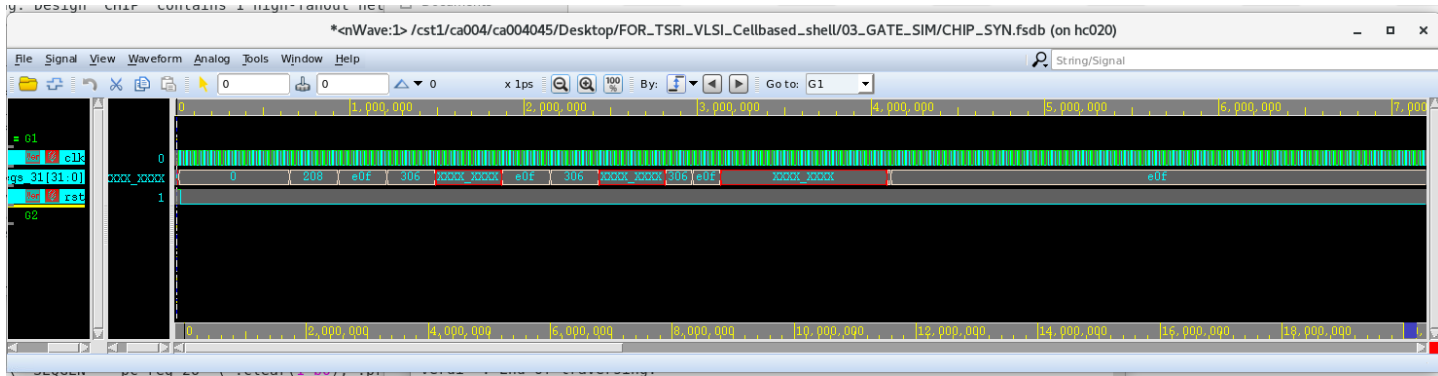
////////////////////////////////////////
// Created by: Synopsys Design Compiler(R)
// Version   : X-2025.06
// Date      : Wed Dec 17 16:48:49 2025
////////////////////////////////////////

module CHIP ( clk, rst, regs_31 );
  output [31:0] regs_31;
  input  clk, rst;
  wire   ipad_clk, ipad_reset;
  wire   [31:0] opad_regs_31;

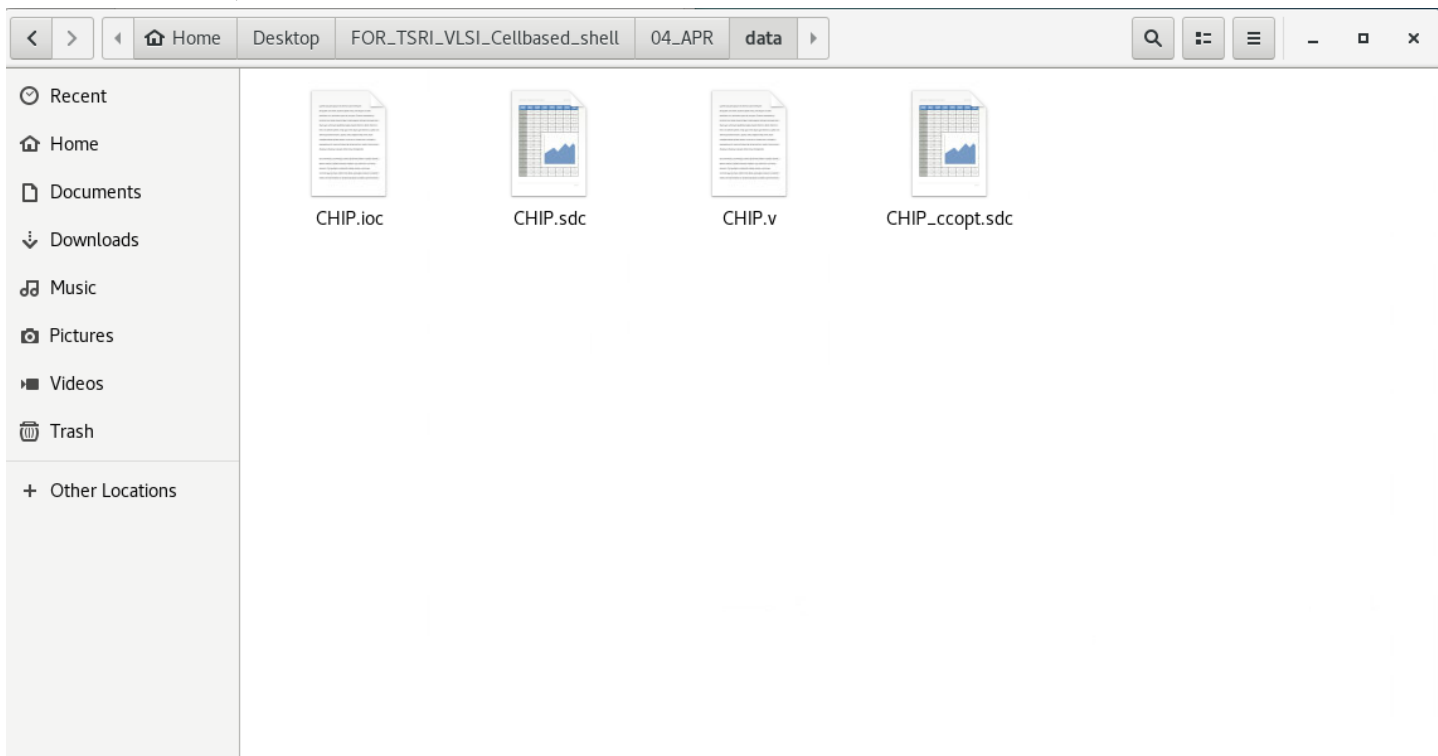
  PDIDGZ_33 u_clk ( .PAD(clk), .C(ipad_clk) );
  PDIDGZ_33 u_reset ( .PAD(rst), .C(ipad_reset) );
  PD016CDG_33 u_regs_31_0 ( .I(opad_regs_31[0]), .PAD(regs_31[0]) );
  PD016CDG_33 u_regs_31_1 ( .I(opad_regs_31[1]), .PAD(regs_31[1]) );
  PD016CDG_33 u_regs_31_2 ( .I(opad_regs_31[2]), .PAD(regs_31[2]) );
  PD016CDG_33 u_regs_31_3 ( .I(opad_regs_31[3]), .PAD(regs_31[3]) );
  PD016CDG_33 u_regs_31_4 ( .I(opad_regs_31[4]), .PAD(regs_31[4]) );
  PD016CDG_33 u_regs_31_5 ( .I(opad_regs_31[5]), .PAD(regs_31[5]) );
  PD016CDG_33 u_regs_31_6 ( .I(opad_regs_31[6]), .PAD(regs_31[6]) );
  PD016CDG_33 u_regs_31_7 ( .I(opad_regs_31[7]), .PAD(regs_31[7]) );
  PD016CDG_33 u_regs_31_8 ( .I(opad_regs_31[8]), .PAD(regs_31[8]) );
  PD016CDG_33 u_regs_31_9 ( .I(opad_regs_31[9]), .PAD(regs_31[9]) );
  PD016CDG_33 u_regs_31_10 ( .I(opad_regs_31[10]), .PAD(regs_31[10]) );
  PD016CDG_33 u_regs_31_11 ( .I(opad_regs_31[11]), .PAD(regs_31[11]) );
  PD016CDG_33 u_regs_31_12 ( .I(opad_regs_31[12]), .PAD(regs_31[12]) );
  PD016CDG_33 u_regs_31_13 ( .I(opad_regs_31[13]), .PAD(regs_31[13]) );
  PD016CDG_33 u_regs_31_14 ( .I(opad_regs_31[14]), .PAD(regs_31[14]) );
  PD016CDG_33 u_regs_31_15 ( .I(opad_regs_31[15]), .PAD(regs_31[15]) );
  PD016CDG_33 u_regs_31_16 ( .I(opad_regs_31[16]), .PAD(regs_31[16]) );
  PD016CDG_33 u_regs_31_17 ( .I(opad_regs_31[17]), .PAD(regs_31[17]) );
  PD016CDG_33 u_regs_31_18 ( .I(opad_regs_31[18]), .PAD(regs_31[18]) );
  PD016CDG_33 u_regs_31_19 ( .I(opad_regs_31[19]), .PAD(regs_31[19]) );
endmodule
```

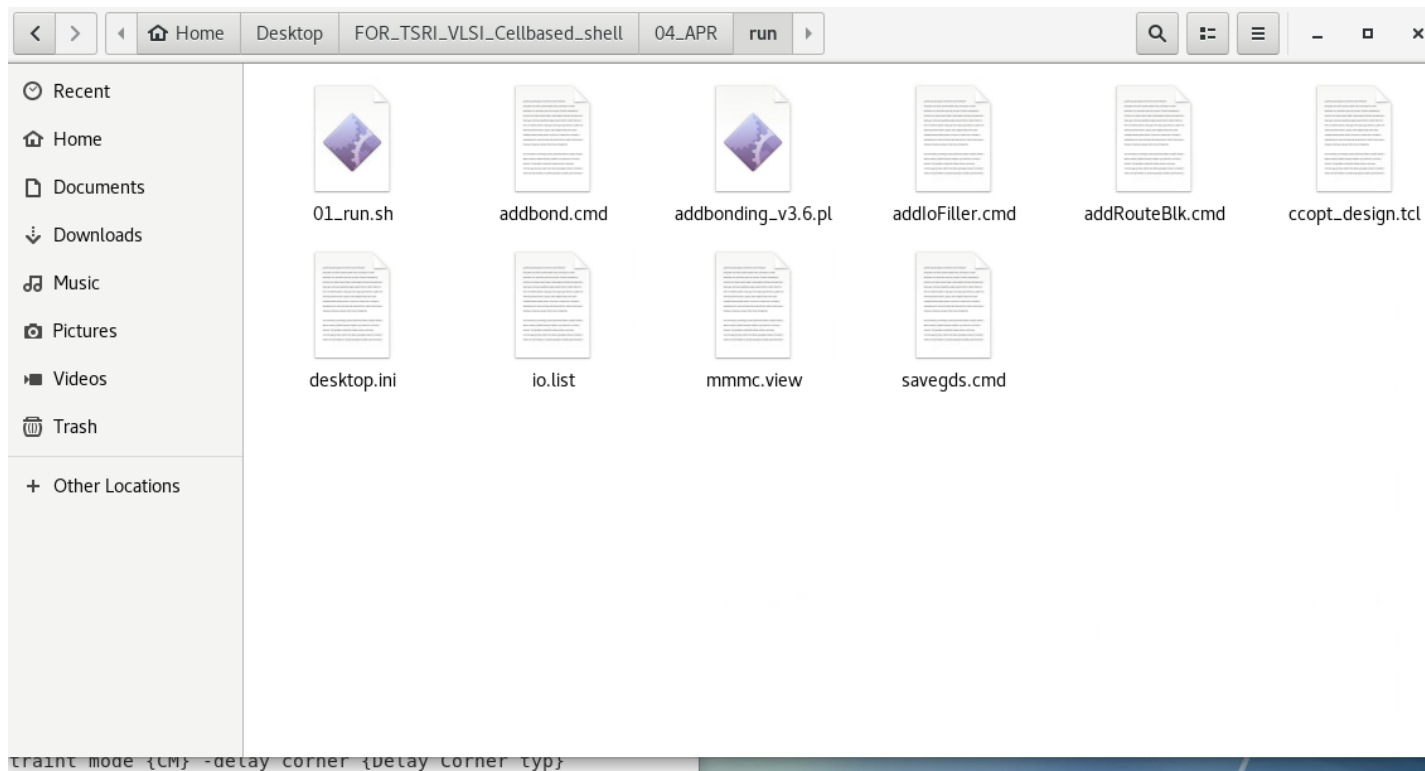
Verilog ▾ Tab Width: 8 ▾ Ln 1, Col 1 ▾ INS

5. Gate-Sim 波形模擬結果：



6. APR 前置準備檔案截圖：





7. APR 設計檔匯入(import design 視窗截圖)：

Design Import — hc083 (on hc083)

Netlist:

☒ Verilog

Files: ...

Top Cell: ☐ Auto Assign ☒ By User:

☐ OA

Library:

Cell:

View:

Technology/Physical Libraries:

☐ OA

Reference Libraries: ...

Abstract View Names:

Layout View Names:

☒ LEF Files

...

Floorplan

IO Assignment File: ...

Power

Power Nets:

Ground Nets:

CPF File: ...

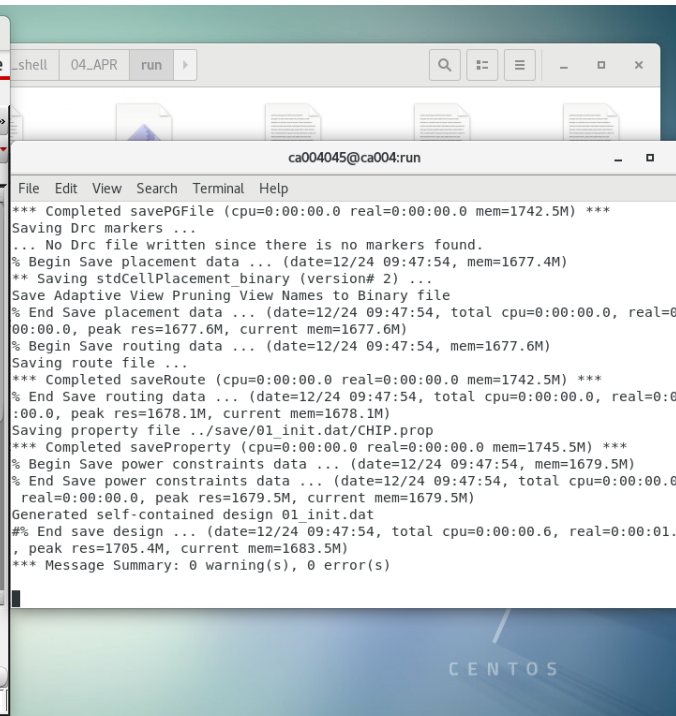
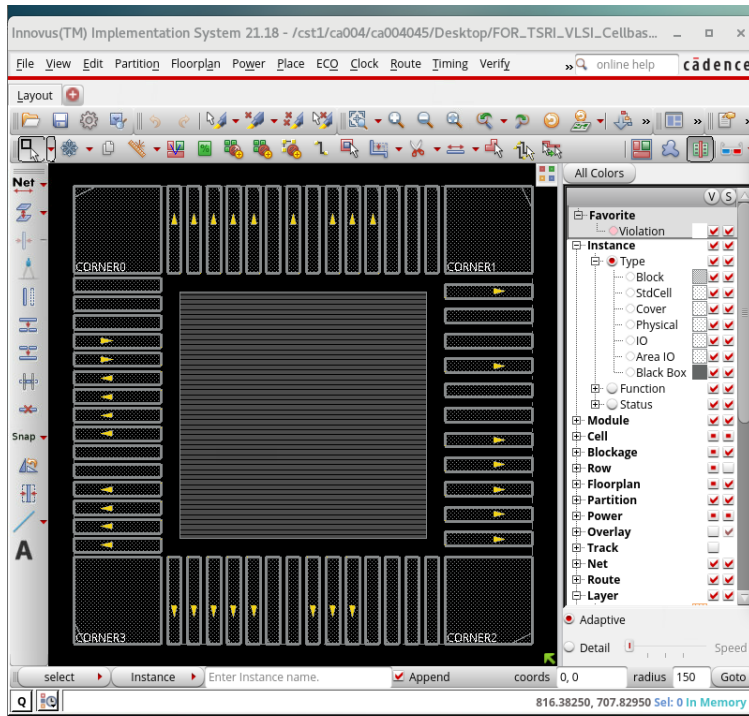
Analysis Configuration

MMMC View Definition File: ...

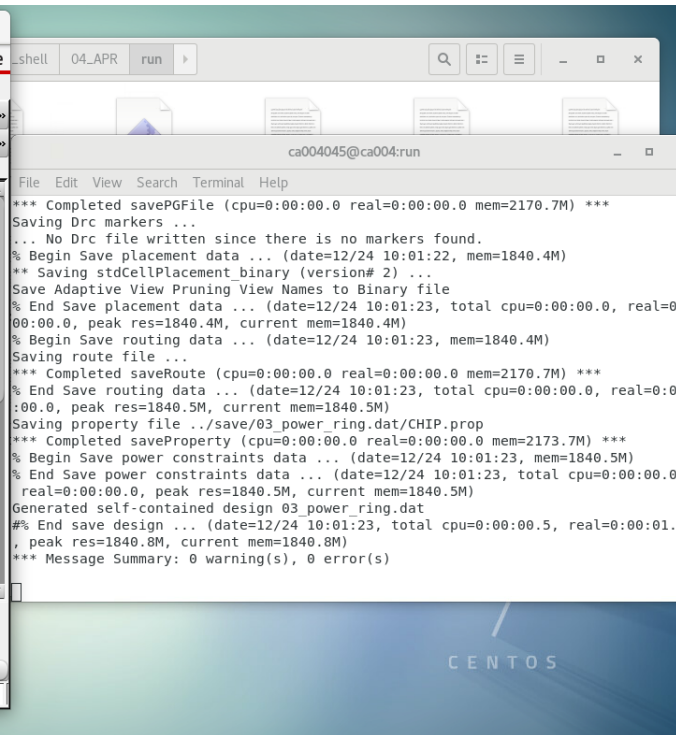
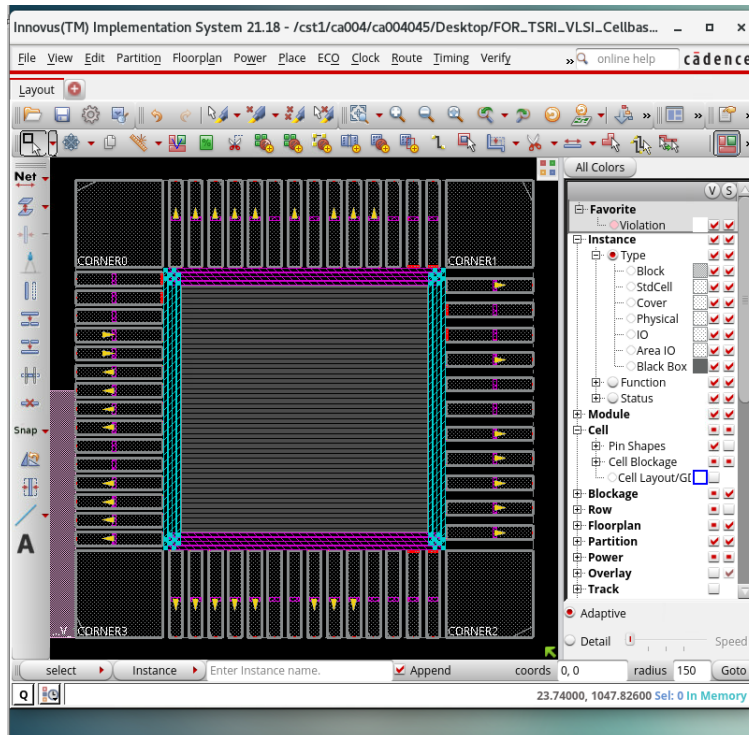
Create Analysis Configuration ...

OK Save... Load... Cancel Help

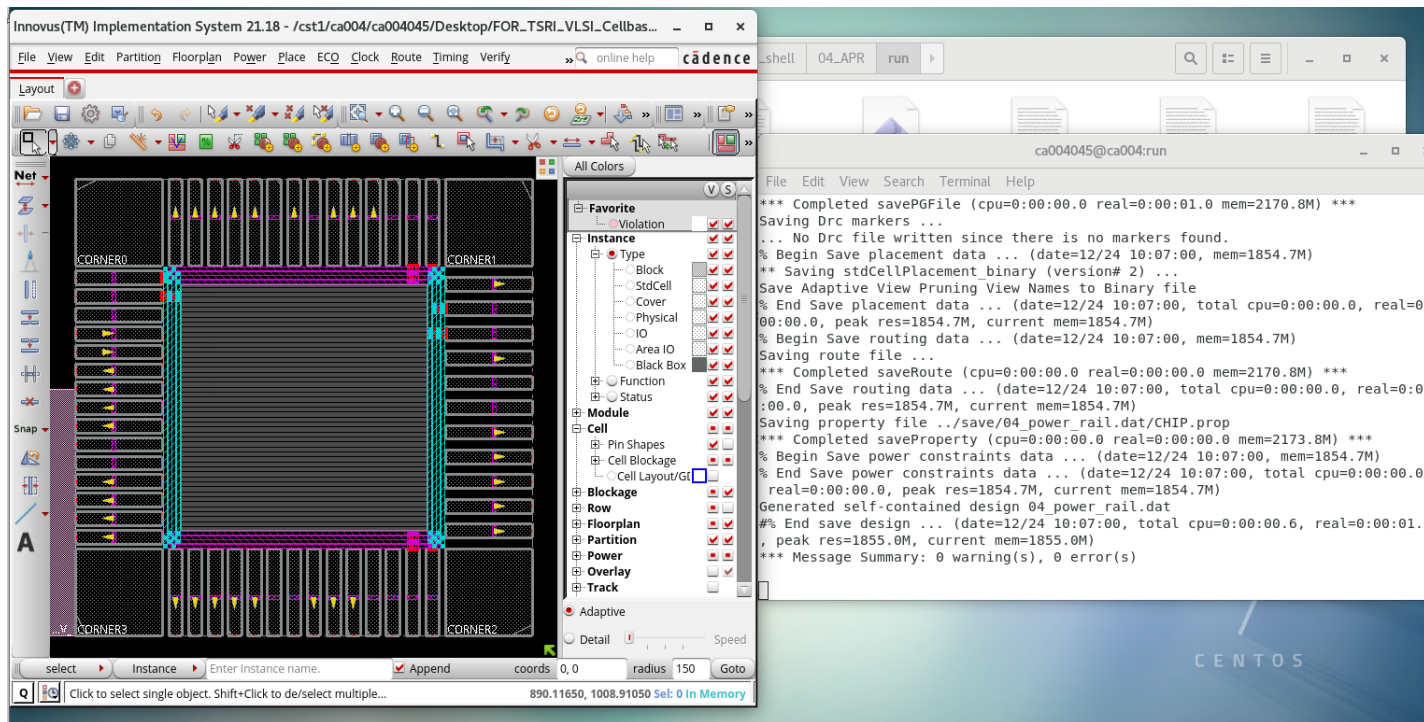
8. APR 執行到 01_init 的結果：



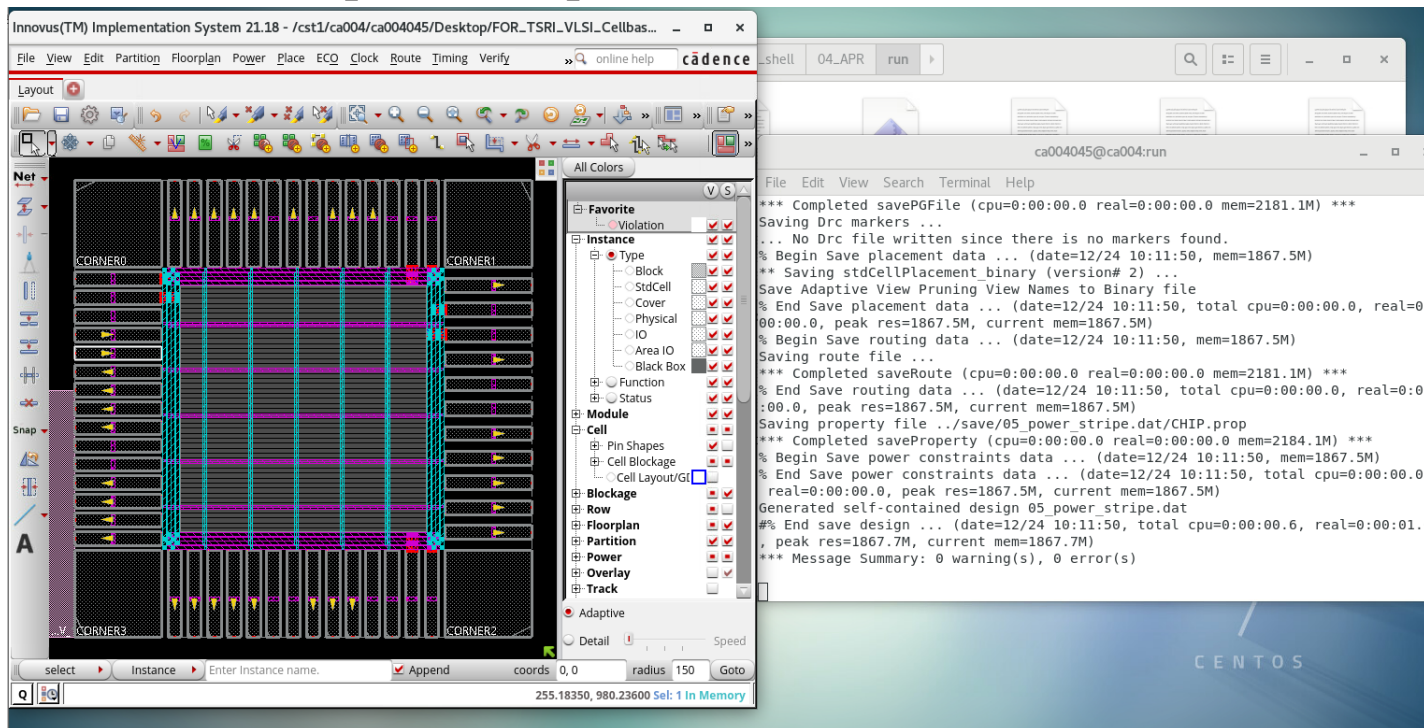
9. APR 執行到 03_power_ring 的結果：



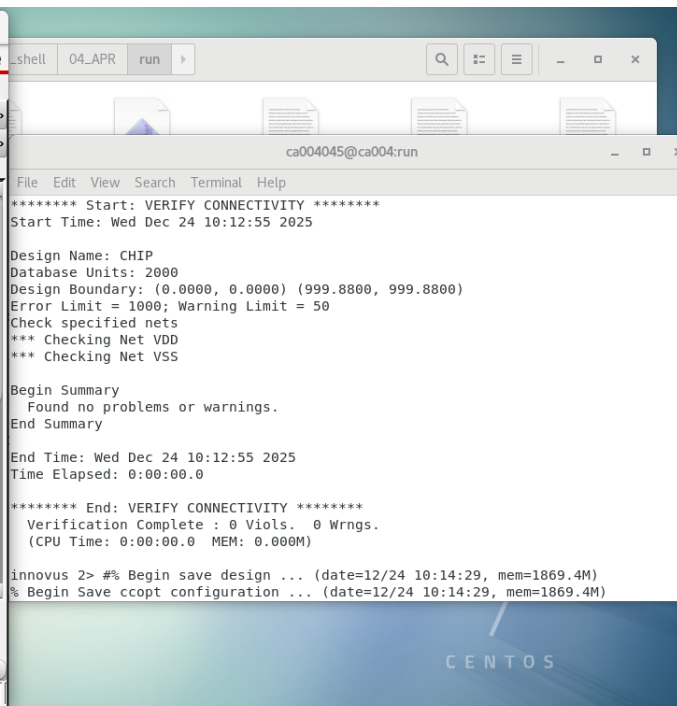
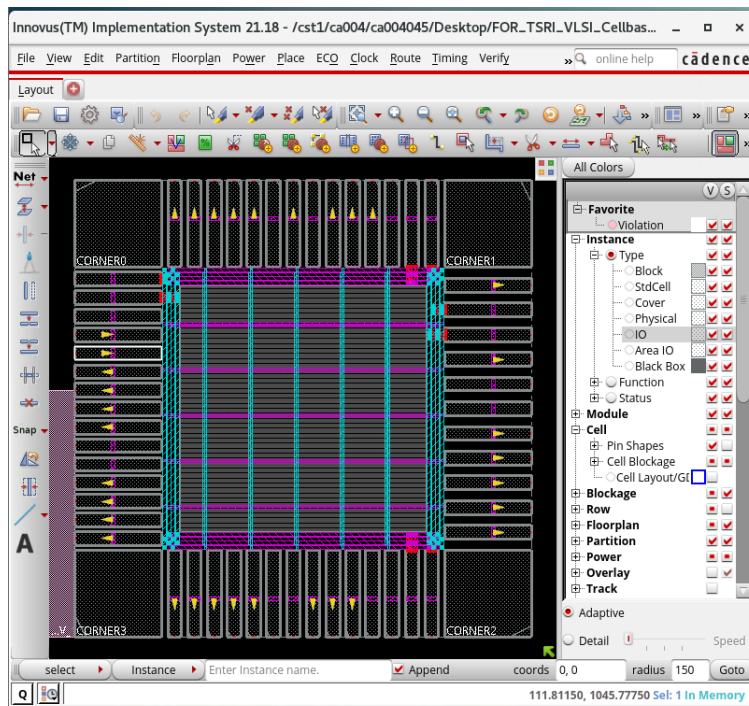
10. APR 執行到 04_power_rail 的結果：



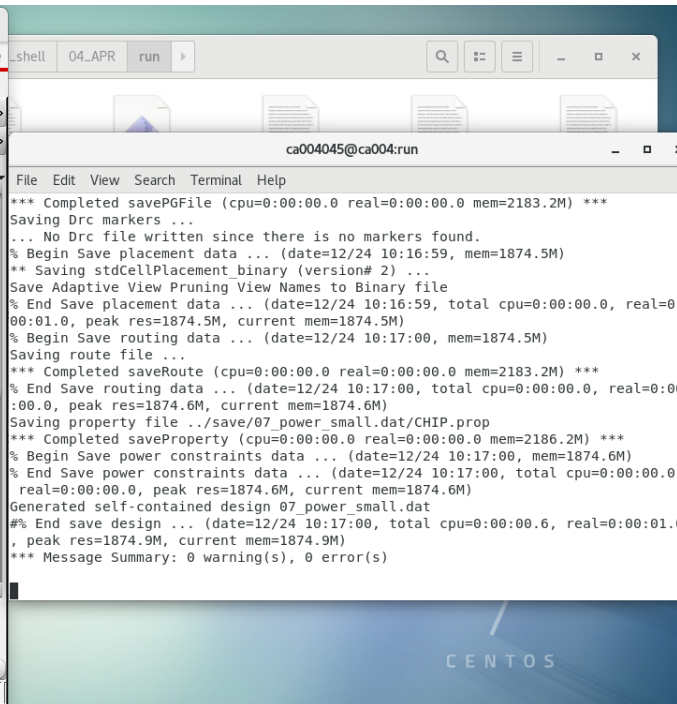
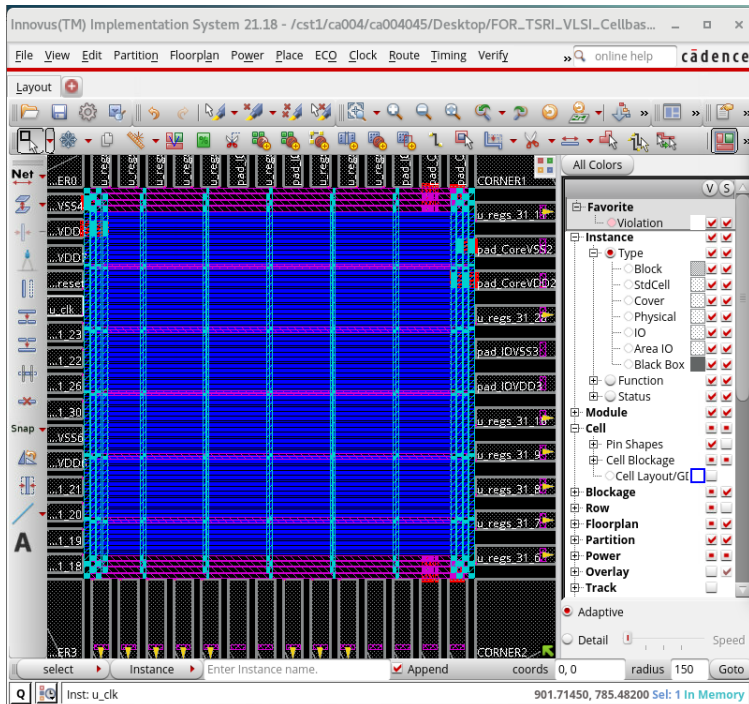
11.APR 執行到 05_power_stripe 的結果：



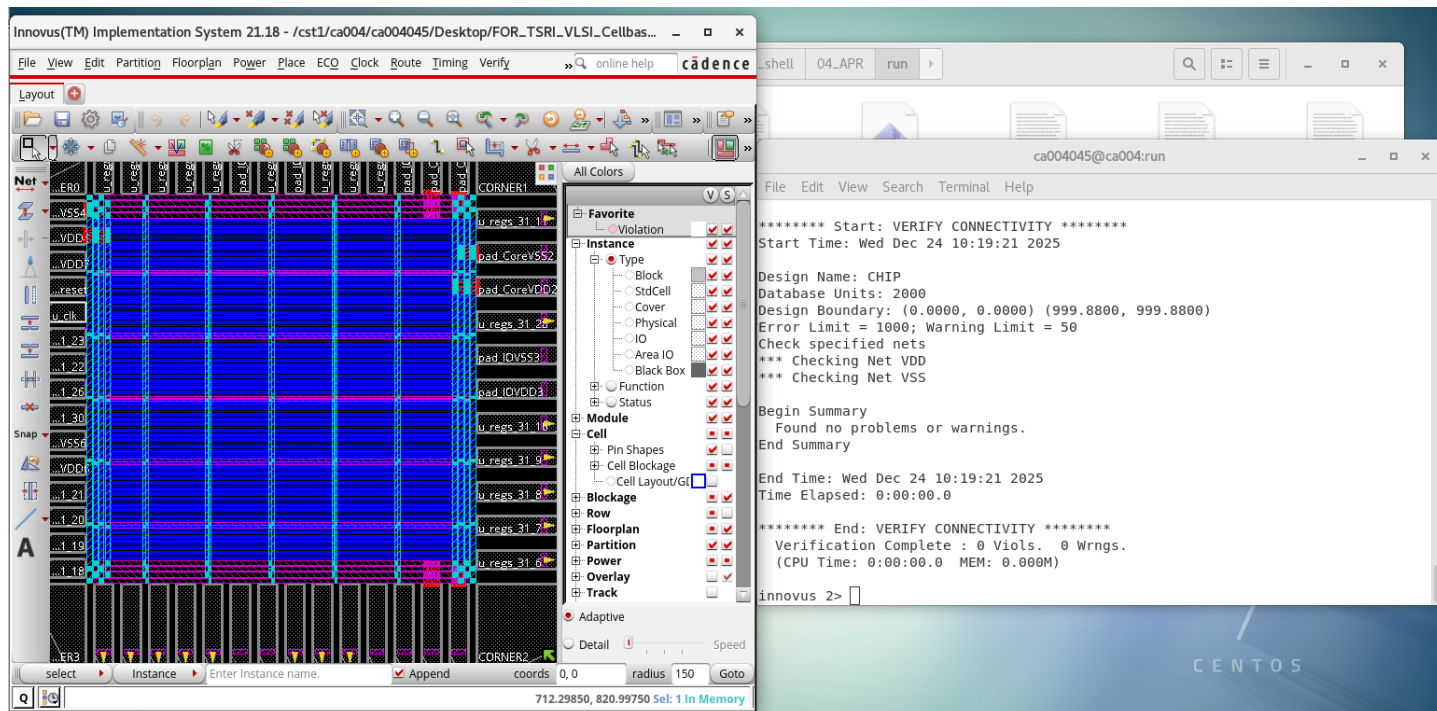
12.APR 執行到 06_verify 的結果：



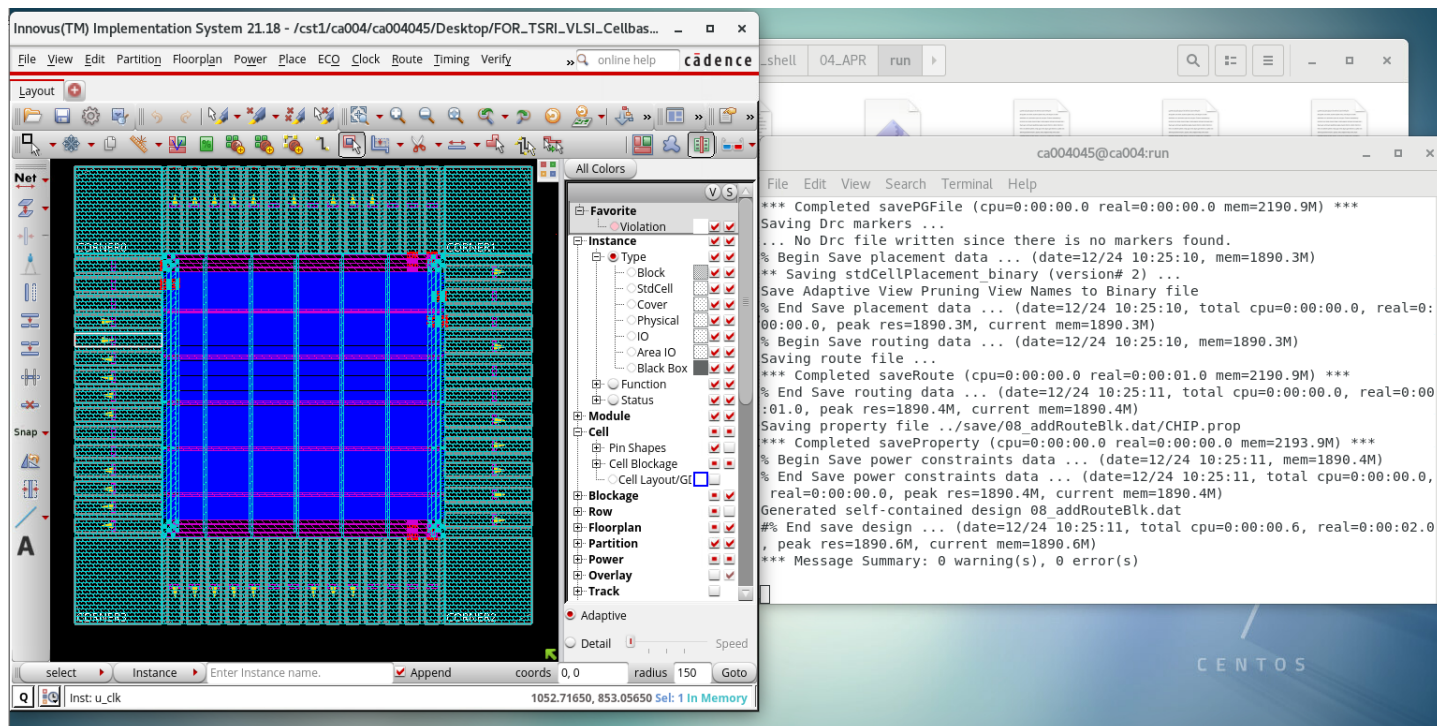
13. APR 執行到 07_power_small 的結果：



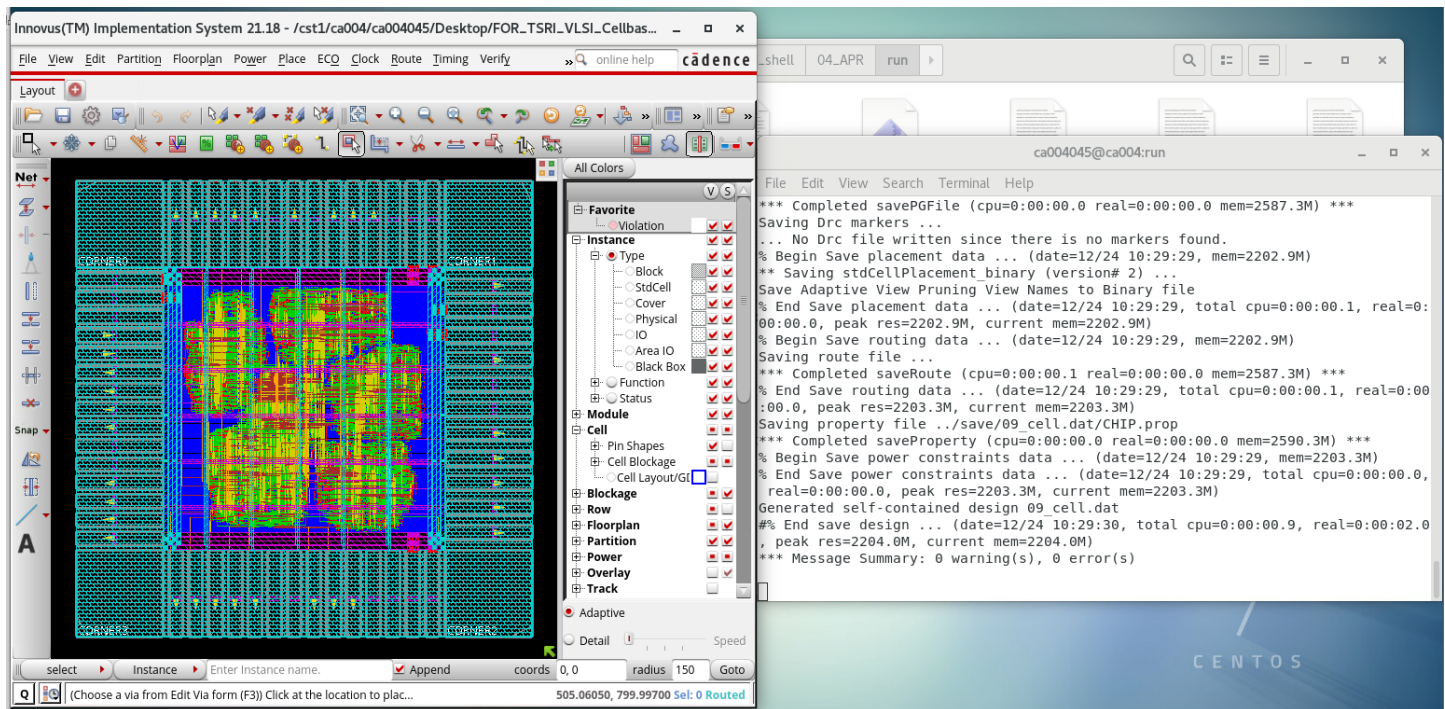
14. 執行完 07_power_small 後的驗證結果(截終端機畫面)：



15. APR 執行到 08_addRouteBlk 的結果：



16. APR 執行到 09_cell 的結果：



17. APR 執行到 10_pre_timing_ok 的結果(截終端機畫面。若有執行最佳化則需要截兩個結果[setup、optimize])：

Setup

Setup views included:
AV_max AV_typ

Setup mode	all	reg2reg	default
WNS (ns):	53.806	80.563	53.806
TNS (ns):	0.000	0.000	0.000
Violating Paths:	0	0	0
All Paths:	5362	5330	1266

DRVs	Real		Total
	Nr nets(terms)	Worst Vio	Nr nets(terms)
max_cap	874 (874)	-0.164	874 (874)
max_tran	711 (15790)	-3.130	743 (15822)
max_fanout	1 (1)	-19	2 (2)
max_length	0 (0)	0	0 (0)

hold

Hold views included:

AV_min

Hold mode	all	reg2reg	default
WNS (ns):	0.006	0.006	40.317
TNS (ns):	0.000	0.000	0.000
Violating Paths:	0	0	0
All Paths:	5362	5330	1266

Density: 49.044%

Routing Overflow: 0.01% H and 0.00% V

Reported timing to dir timingReports

Total CPU time: 2.08 sec

Total Real time: 2.0 sec

Total Memory Usage: 2667.808594 Mbytes

*** timeDesign #2 [finish] : cpu/real = 0:00:02.1/0:00:02.1 (1.0), totSession c
u/real = 0:03:03.0/0:55:24.9 (0.1), mem = 2667.8M

innovus 7> █

optimize

ca004045@ca004:run

File Edit View Search Terminal Help

Setup views included:

AV_max AV_typ

Setup mode	all	reg2reg	default
WNS (ns):	56.112	87.685	56.112
TNS (ns):	0.000	0.000	0.000
Violating Paths:	0	0	0
All Paths:	5362	5330	1266

DRVs	Real		Total
	Nr nets(terms)	Worst Vio	Nr nets(terms)
max_cap	0 (0)	0.000	0 (0)
max_tran	0 (0)	0.000	32 (32)
max_fanout	1 (1)	-19	2 (2)
max_length	0 (0)	0	0 (0)

18. APR 執行到 12_post_timing_ok 的結果(截終端機畫面。若有執行最佳化則需要截三個結果[setup、hold、optimize])：

setup

ca004045@ca004:run			
File Edit View Search Terminal Help			
Setup mode	all	reg2reg	default
WNS (ns):	55.016	87.856	55.016
TNS (ns):	0.000	0.000	0.000
Violating Paths:	0	0	0
All Paths:	5362	5330	1266
DRVs	Real		Total
	Nr nets(terms)	Worst Vio	Nr nets(terms)
max_cap	0 (0)	0.000	0 (0)
max_tran	0 (0)	0.000	32 (32)
max_fanout	1 (1)	-19	2 (2)
max_length	0 (0)	0	0 (0)
Density: 49.292%			
Routing Overflow: 0.01% H and 0.00% V			

Reported timing to dir timingReports			

hold

File Edit View Search Terminal Help

Setup views included:

AV_max AV_typ

Setup mode	all	reg2reg	default
WNS (ns):	55.015	87.826	55.015
TNS (ns):	0.000	0.000	0.000
Violating Paths:	0	0	0
All Paths:	5362	5330	1266

DRVs	Real		Total
	Nr nets(terms)	Worst Vio	Nr nets(terms)
max_cap	0 (0)	0.000	0 (0)
max_tran	0 (0)	0.000	32 (32)
max_fanout	0 (0)	0	1 (1)
max_length	0 (0)	0	0 (0)

Density: 49.295%

File Edit View Search Terminal Help

Hold mode	all	reg2reg	default
WNS (ns):	0.103	0.103	39.846
TNS (ns):	0.000	0.000	0.000
Violating Paths:	0	0	0
All Paths:	5362	5330	1266

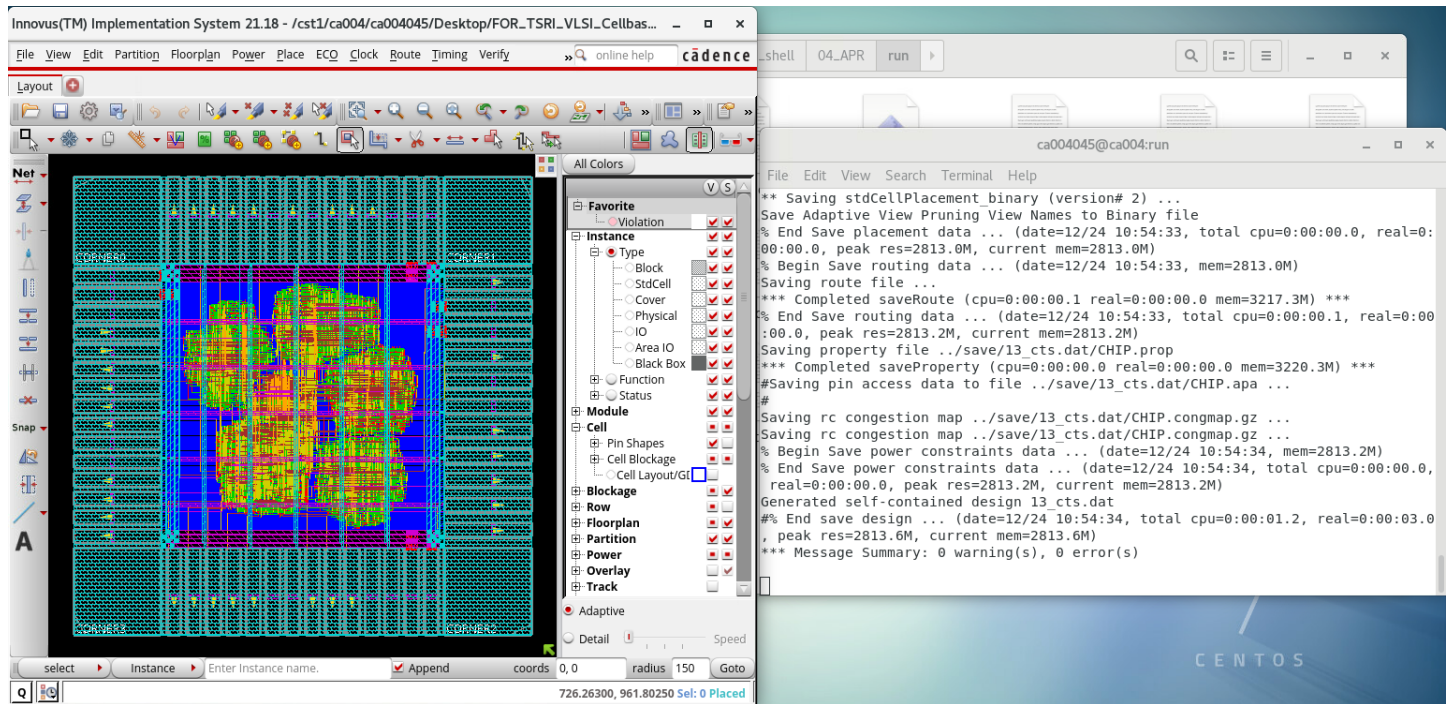
DRVs	Real		Total
	Nr nets(terms)	Worst Vio	Nr nets(terms)
max_cap	0 (0)	0.000	0 (0)
max_tran	0 (0)	0.000	32 (32)
max_fanout	0 (0)	0	1 (1)
max_length	0 (0)	0	0 (0)

Density: 49.295%

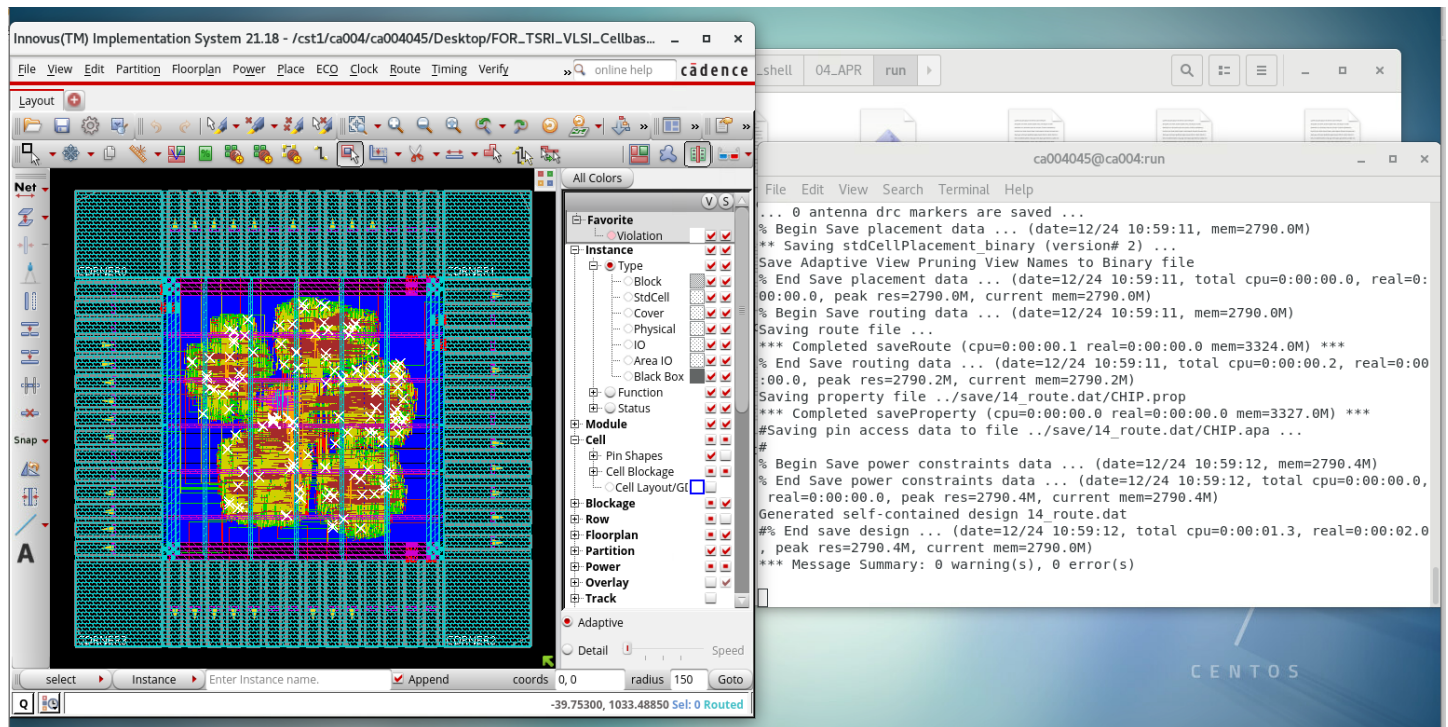
Routing Overflow: 0.01% H and 0.00% V

**optDesign ... cpu = 0:00:11, real = 0:00:11, mem = 2793.2M, totSessionCpu=0:09

19. APR 執行到 13_CTS 的結果：



20. APR 執行到 14_route 的結果：



21. APR 執行到 15_route_timing_ok 的結果(截終端機畫面。若有執行最佳化則需要截三個結果[setup、hold、optimize])：

setup

ca004045@ca004:run				
File Edit View Search Terminal Help				
+-----+-----+-----+-----+				
Setup mode		all	reg2reg	default
+-----+-----+-----+-----+				
WNS (ns):		55.060	88.035	55.060
TNS (ns):		0.000	0.000	0.000
Violating Paths:		0	0	0
All Paths:		5362	5330	1266
+-----+-----+-----+-----+				
+-----+-----+-----+-----+				
DRVs		Real		Total
		+-----+-----+		+-----+
		Nr nets(terms)	Worst Vio	Nr nets(terms)
+-----+-----+-----+-----+				
max_cap		0 (0)	0.000	0 (0)
max_tran		0 (0)	0.000	32 (32)
max_fanout		0 (0)	0	1 (1)
max_length		0 (0)	0	0 (0)
+-----+-----+-----+-----+				
Density: 49.492%				

Reported timing to dir timingReports				
Total CPU time: 9.4 sec				

hold

22. APR 執行到 16_sign_off_timing_ok 的結果(截終端機畫面。若有執行最佳化則需要截三個結果[setup、hold、optimize])：

setup

ca004045@ca004:run

File Edit View Search Terminal Help

timeDesign Summary

Setup views included:
AV_max AV_typ

Setup mode	all	reg2reg	default
WNS (ns):	54.980	87.584	54.980
TNS (ns):	0.000	0.000	0.000
Violating Paths:	0	0	0
All Paths:	5362	5330	1266

DRVs	Real		Total
	Nr nets(terms)	Worst Vio	Nr nets(terms)
max_cap	0 (0)	0.000	0 (0)
max_tran	0 (0)	0.000	32 (32)
max_fanout	0 (0)	0	1 (1)

hold

```
ca004045@ca004:run
File Edit View Search Terminal Help
Hold views included:
AV_min

+-----+-----+-----+
| Hold mode | all | reg2reg | default |
+-----+-----+-----+
| WNS (ns): | 0.105 | 0.105 | 39.844 |
| TNS (ns): | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 0 | 0 | 0 |
| All Paths: | 5362 | 5330 | 1266 |
+-----+-----+-----+

Density: 49.493%
(100.000% with Fillers)
-----
Total CPU time: 20.22 sec
Total Real time: 20.0 sec
Total Memory Usage: 4171.535156 Mbytes
Reset AAE Options
Message <GLOBAL-100> has exceeded the message display limit of '20'. Use 'set_m
ssage -no_limit -id list_of_msgIDs' to reset the message limit.
*** timeDesign #10 [finish] : cpu/real = 0:00:20.2/0:00:20.2 (1.0), totSession
pu/real = 0:17:20.3/1:45:11.9 (0.2), mem = 4171.5M
innovus 11>
```

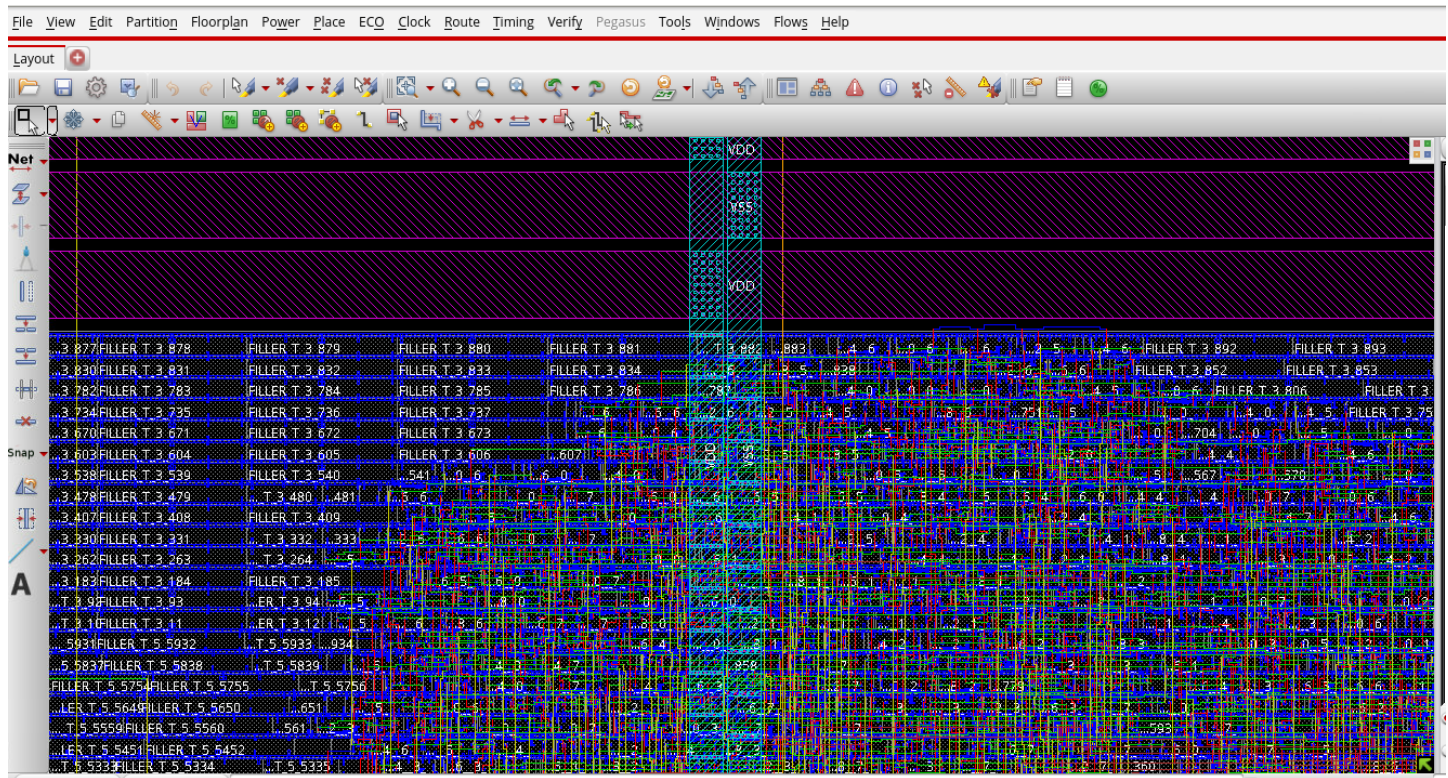
optimize

```
ca004045@ca004:run
File Edit View Search Terminal Help
Setup views included:
AV_max AV_typ
Hold views included:
AV_min

+-----+-----+-----+
| Setup mode | all | reg2reg | default |
+-----+-----+-----+
| WNS (ns): | 55.060 | 88.035 | 55.060 |
| TNS (ns): | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 0 | 0 | 0 |
| All Paths: | 5362 | 5330 | 1266 |
+-----+-----+-----+

+-----+-----+-----+
| Hold mode | all | reg2reg | default |
+-----+-----+-----+
| WNS (ns): | 0.103 | 0.103 | 39.836 |
| TNS (ns): | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 0 | 0 | 0 |
| All Paths: | 5362 | 5330 | 1266 |
+-----+-----+-----+
+-----+-----+-----+
```

23. APR 執行到 17_core_filler 的結果(放大到可以看到 filler 後再截圖)：



24. APR 執行到 18_verify 的結果(截終端機畫面)：

```
ca004045@ca004:run
File Edit View Search Terminal Help

Design Name: CHIP
Database Units: 2000
Design Boundary: (0.0000, 0.0000) (999.8800, 999.8800)
Error Limit = 1000; Warning Limit = 50
Check all nets
**** 11:30:57 **** Processed 5000 nets.
**** 11:30:57 **** Processed 10000 nets.
**** 11:30:57 **** Processed 15000 nets.
Net_NULL: has special routes with opens.

Begin Summary
  7 Problem(s) (IMPVFC-200): Special Wires: Pieces of the net are not connecte
d together.
  7 total info(s) created.
End Summary

End Time: Wed Dec 24 11:30:57 2025
Time Elapsed: 0:00:00.0

***** End: VERIFY CONNECTIVITY *****
Verification Complete : 7 Viols.  0 Wrngs.
(CPU Time: 0:00:00.6  MEM: -0.648M)
```

25. APR 執行到 19_dummy 的結果：

The screenshot displays the Cadence Innovus Implementation System interface. The main window shows a netlist view with a grid of cells and connections. The right-hand pane shows a list of components and their properties, including 'Block', 'StdCell', 'Cover', 'Physical', 'IO', 'Area IO', 'Black Box', 'Function', 'Status', 'Module', 'Cell', 'Pin Shapes', 'Cell Blockage', 'Cell Layout/Gt', 'Blockage', 'Row', 'Floorplan', 'Partition', 'Power', 'Overlay', 'Track', and 'Adaptive'. The bottom status bar indicates 'select Instance Enter Instance name. Append coords 0,0 radius 150 Goto'. To the right, a terminal window shows the output of the APR (Automated Placement and Routing) process for 19_dummy. The output includes various status messages, timing information, and resource usage statistics.

```
... No Drc file written since there is no markers found.
% Begin Save placement data ... (date=12/24 11:38:54, mem=4079.9M)
** Saving stdCellPlacement binary (version# 2) ...
Save Adaptive View Pruning View Names to Binary file
% End Save placement data ... (date=12/24 11:38:54, total cpu=0:00:00.1, real=0:00:00.0, peak res=4079.9M, current mem=4079.9M)
% Begin Save routing data ... (date=12/24 11:38:54, mem=4079.9M)
Saving route file ...
*** Completed saveRoute (cpu=0:00:00.2 real=0:00:01.0 mem=4714.1M) ***
% End Save routing data ... (date=12/24 11:38:55, total cpu=0:00:00.2, real=0:00:01.0, peak res=4080.1M, current mem=4080.1M)
Saving property file ../save/19_dummy.dat/CHIP.prop
*** Completed saveProperty (cpu=0:00:00.0 real=0:00:00.0 mem=4717.1M) ***
#Saving pin access data to file ../save/19_dummy.dat/CHIP.apa ...
#
% Begin Save power constraints data ... (date=12/24 11:38:55, mem=4080.1M)
% End Save power constraints data ... (date=12/24 11:38:55, total cpu=0:00:00.0, real=0:00:00.0, peak res=4080.1M, current mem=4080.1M)
Generated self-contained design 19_dummy.dat
% End save design ... (date=12/24 11:38:55, total cpu=0:00:01.5, real=0:00:03.0, peak res=4087.3M, current mem=4080.3M)
*** Message Summary: 0 warning(s), 0 error(s)
```

26. APR 執行到 20_verify 的結果(截終端機畫面)：

File Edit View Search Terminal Help

```
Design Name: CHIP
Database Units: 2000
Design Boundary: (0.0000, 0.0000) (999.8800, 999.8800)
Error Limit = 1000; Warning Limit = 50
Check all nets
**** 11:39:39 **** Processed 5000 nets.
**** 11:39:39 **** Processed 10000 nets.
**** 11:39:39 **** Processed 15000 nets.
Net _NULL: has special routes with opens.

Begin Summary
  7 Problem(s) (IMPVFC-200): Special Wires: Pieces of the net are not connecte
d together.
  7 total info(s) created.
End Summary

End Time: Wed Dec 24 11:39:40 2025
Time Elapsed: 0:00:01.0

***** End: VERIFY CONNECTIVITY *****
Verification Complete : 7 Viols.  0 Wrngs.
(CPU Time: 0:00:00.6  MEM: -0.234M)
```

innovus 13> █

27.執行天線效應驗證(Antenna)的結果(截終端機畫面)：

File Edit View Search Terminal Help

```
  7 Problem(s) (IMPVFC-200): Special Wires: Pieces of the net are not connecte
d together.
  7 total info(s) created.
End Summary

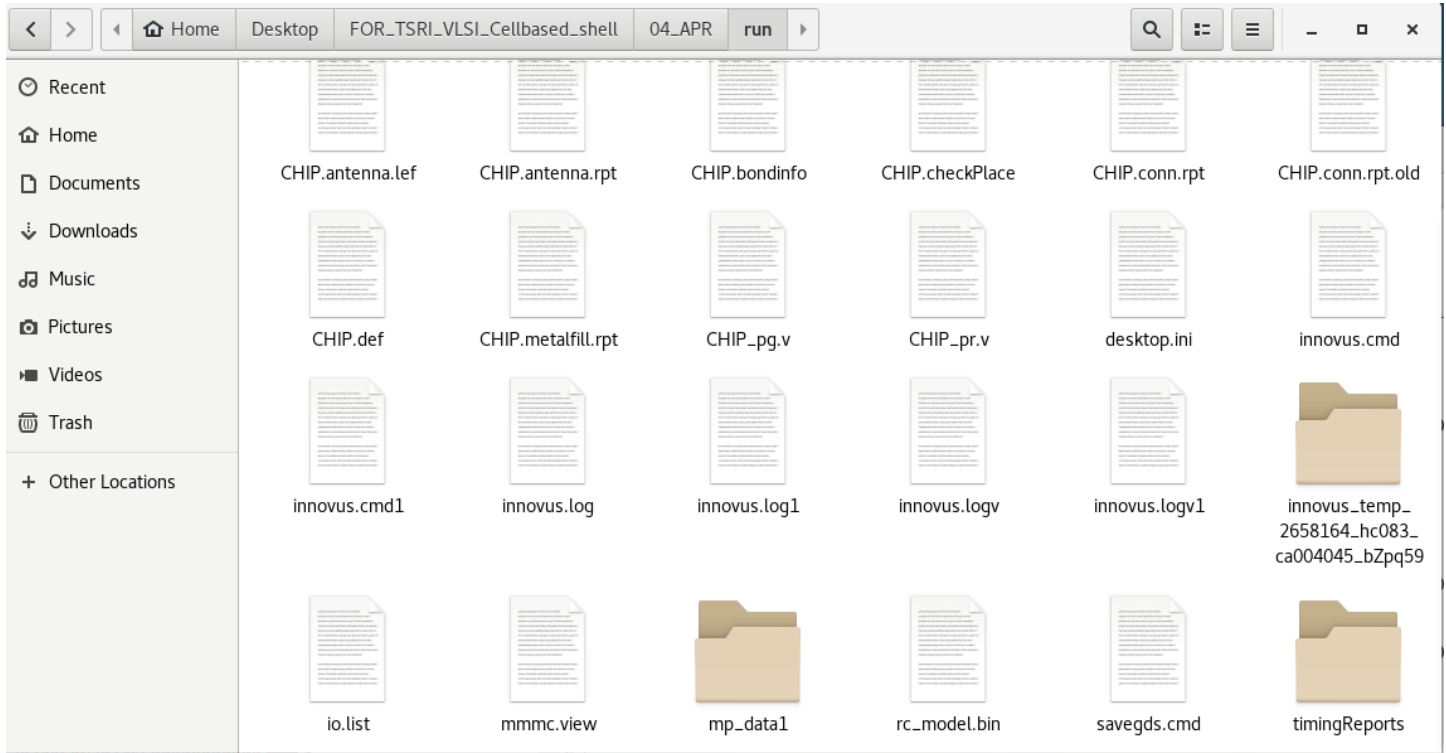
End Time: Wed Dec 24 11:39:40 2025
Time Elapsed: 0:00:01.0

***** End: VERIFY CONNECTIVITY *****
Verification Complete : 7 Viols.  0 Wrngs.
(CPU Time: 0:00:00.6  MEM: -0.234M)

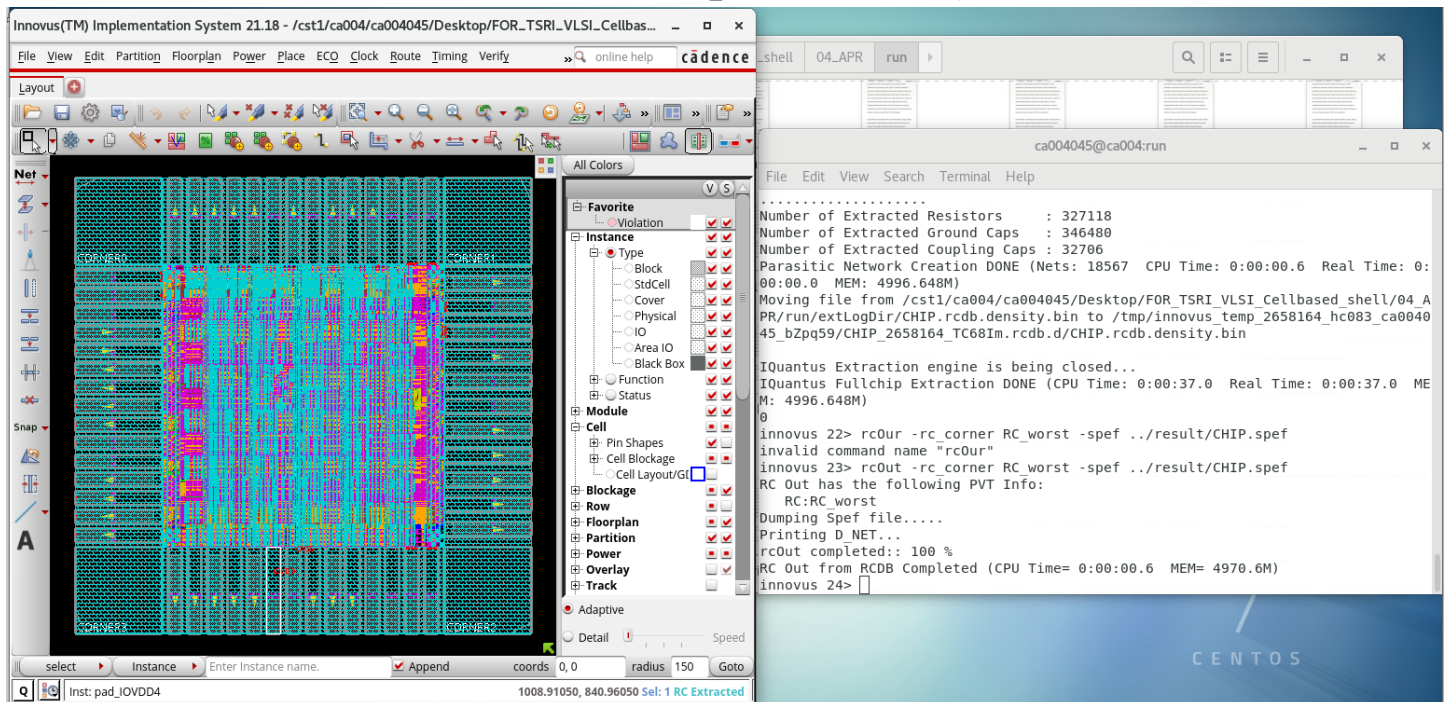
innovus 13>
***** START VERIFY ANTENNA *****
Report File: CHIP.antenna.rpt
LEF Macro File: CHIP.antenna.lef
5000 nets processed: 0 violations
10000 nets processed: 0 violations
15000 nets processed: 0 violations
Verification Complete: 0 Violations
***** DONE VERIFY ANTENNA *****
(CPU Time: 0:00:00.7  MEM: 0.000M)

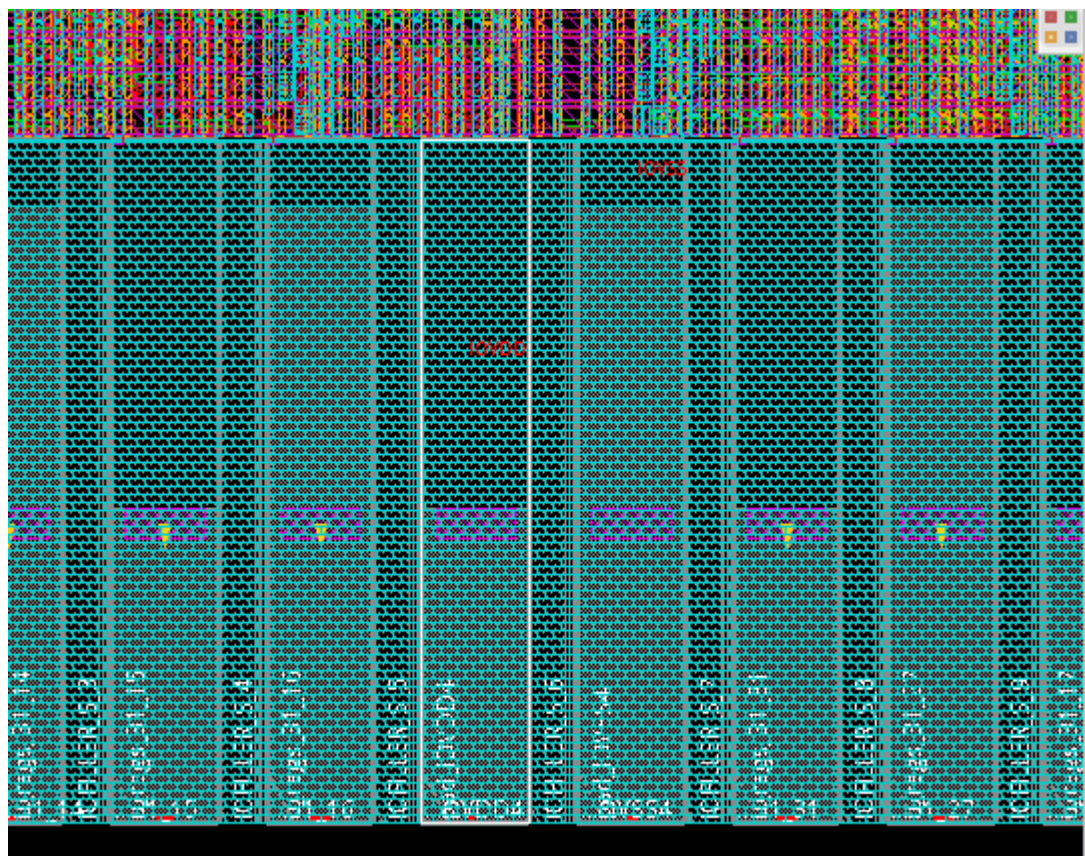
innovus 13> █
```

28. 輸出 CHIP_pr.v 和 CHIP_pg.v 的結果(截資料夾內的 icon)：

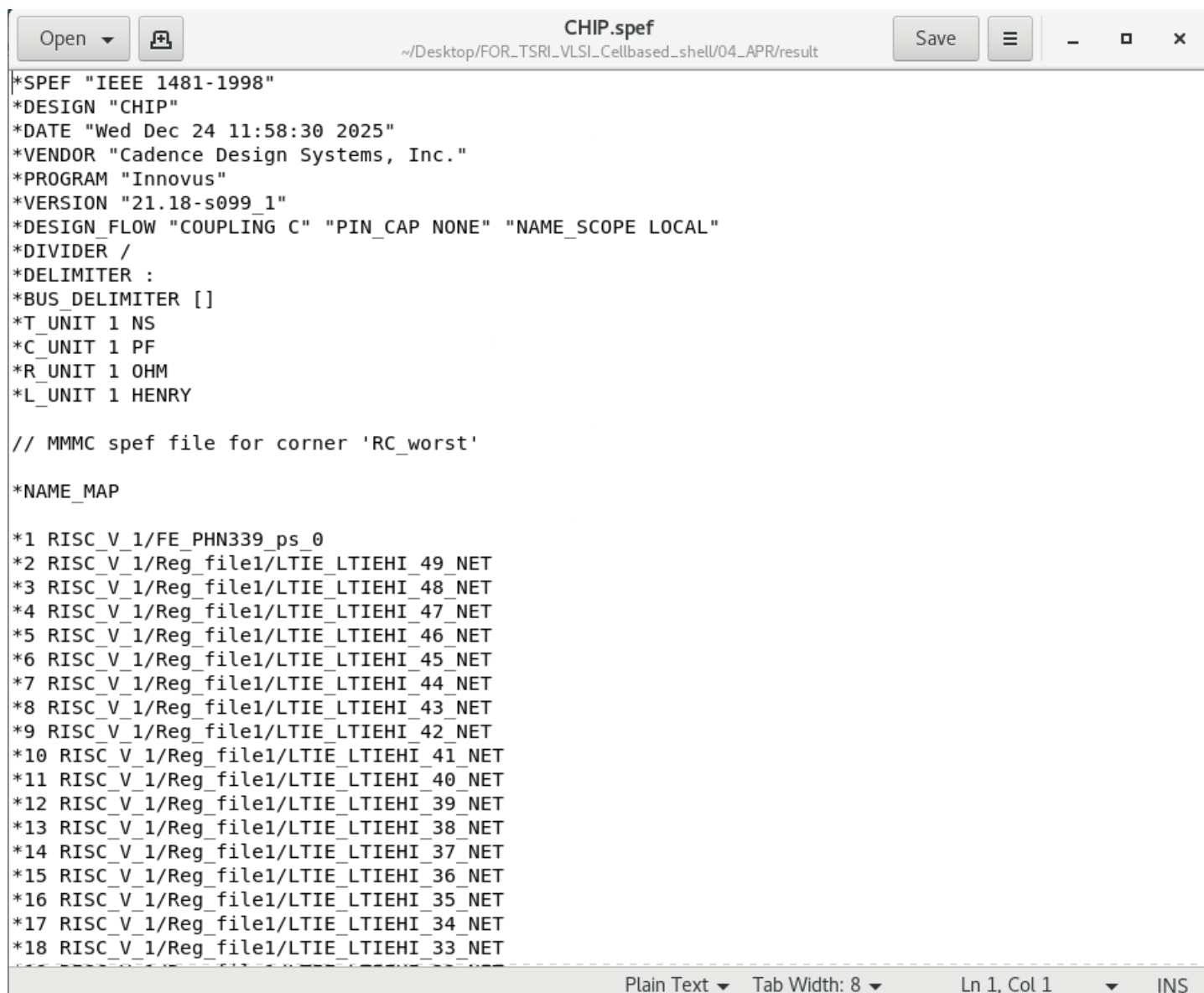


29. APR 執行到 22_text_ok 的結果(除了截整個畫面外，也要放大截 IOVDD 和 IOVSS[要連 IO pad 一起截])：





30.APR 執行到 RC Extraction 的結果 (截圖 CHIP.spef 的內容)



The screenshot shows a text editor window titled "CHIP.spf" with a file path of "~/Desktop/FOR_TSRI_VLSI_Cellbased_shell/04_APR/result". The editor contains a .spf file with the following content:

```
*SPEF "IEEE 1481-1998"
*DESIGN "CHIP"
*DATE "Wed Dec 24 11:58:30 2025"
*VENDOR "Cadence Design Systems, Inc."
*PROGRAM "Innovus"
*VERSION "21.18-s099_1"
*DESIGN_FLOW "COUPLING C" "PIN_CAP NONE" "NAME_SCOPE LOCAL"
*DIVIDER /
*DELIMITER :
*BUS_DELIMITER []
*T_UNIT 1 NS
*C_UNIT 1 PF
*R_UNIT 1 OHM
*L_UNIT 1 HENRY

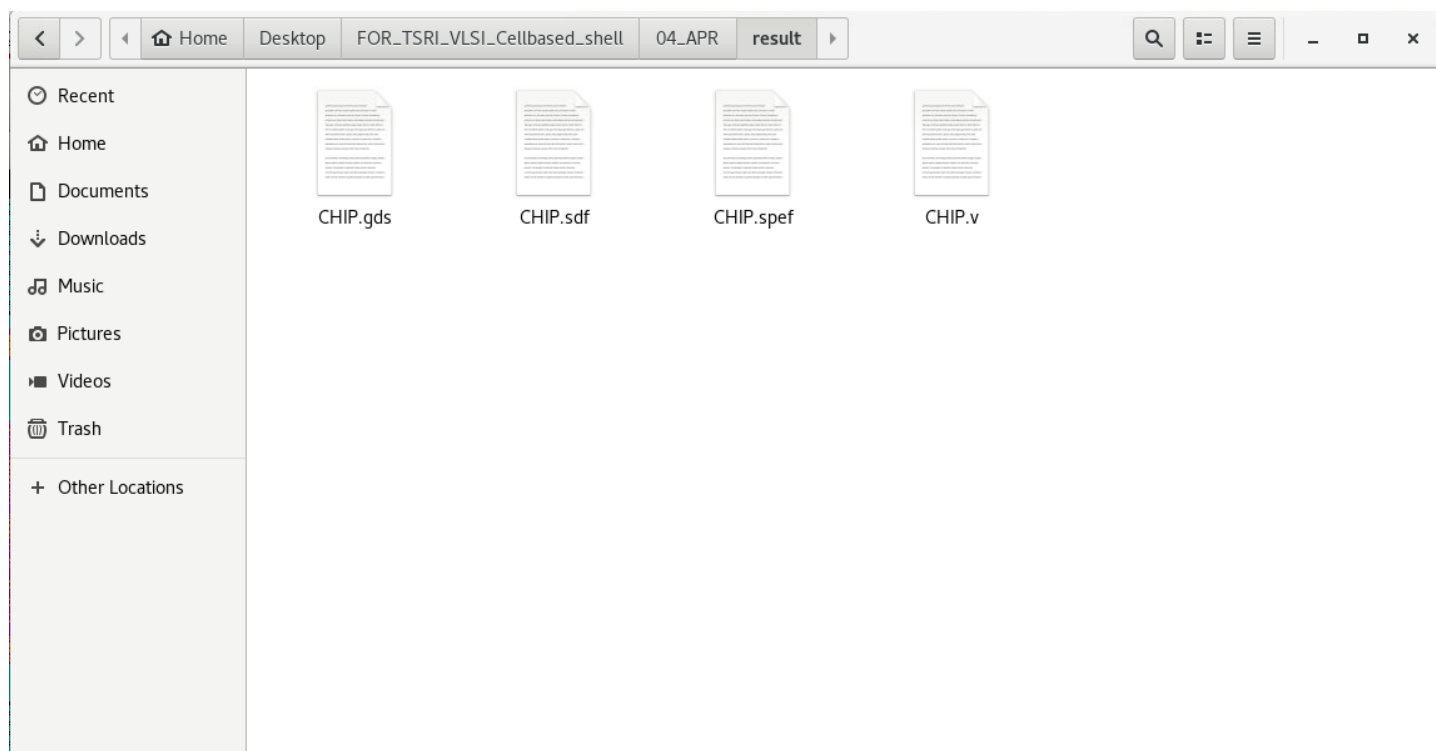
// MMMC spf file for corner 'RC_worst'

*NAME_MAP

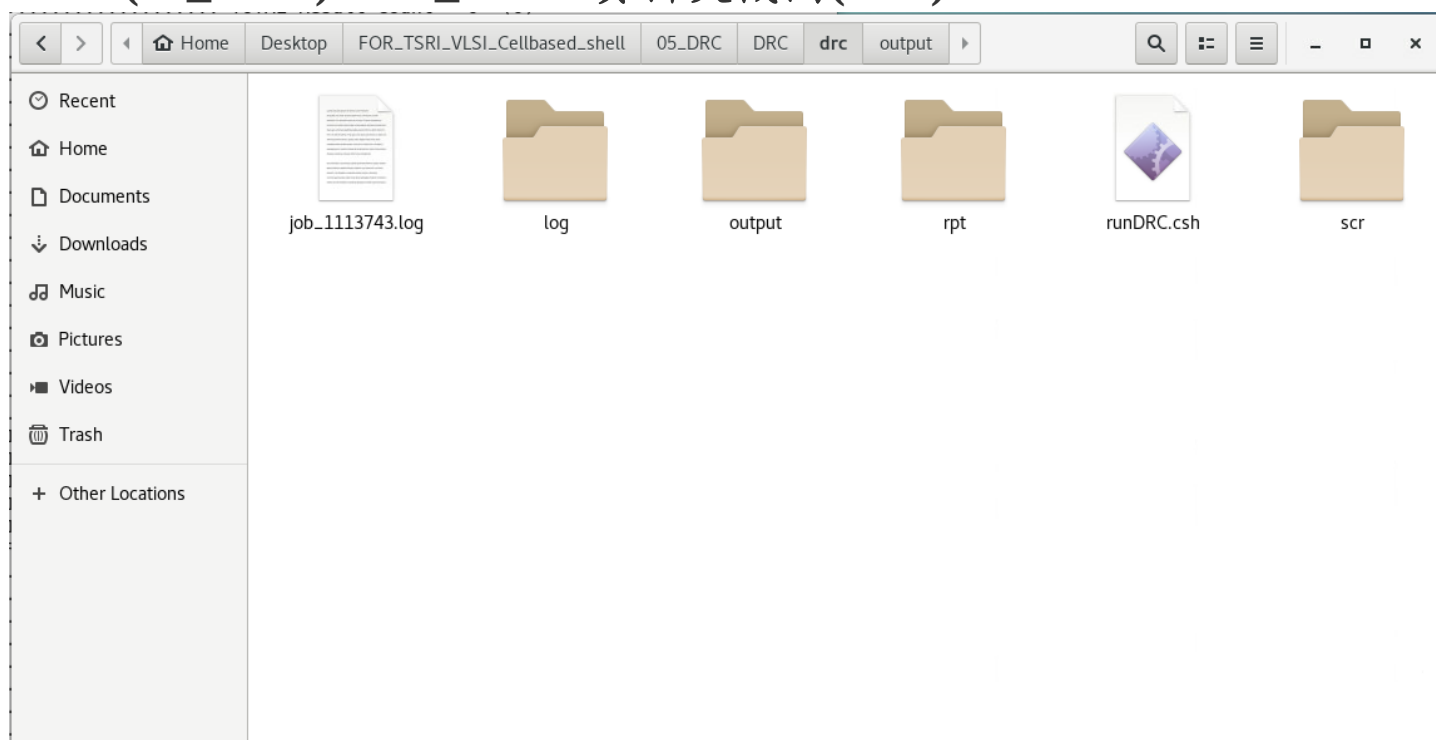
*1 RISC_V_1/FE_PHN339_ps_0
*2 RISC_V_1/Reg_file1/LTIE_LTIEHI_49_NET
*3 RISC_V_1/Reg_file1/LTIE_LTIEHI_48_NET
*4 RISC_V_1/Reg_file1/LTIE_LTIEHI_47_NET
*5 RISC_V_1/Reg_file1/LTIE_LTIEHI_46_NET
*6 RISC_V_1/Reg_file1/LTIE_LTIEHI_45_NET
*7 RISC_V_1/Reg_file1/LTIE_LTIEHI_44_NET
*8 RISC_V_1/Reg_file1/LTIE_LTIEHI_43_NET
*9 RISC_V_1/Reg_file1/LTIE_LTIEHI_42_NET
*10 RISC_V_1/Reg_file1/LTIE_LTIEHI_41_NET
*11 RISC_V_1/Reg_file1/LTIE_LTIEHI_40_NET
*12 RISC_V_1/Reg_file1/LTIE_LTIEHI_39_NET
*13 RISC_V_1/Reg_file1/LTIE_LTIEHI_38_NET
*14 RISC_V_1/Reg_file1/LTIE_LTIEHI_37_NET
*15 RISC_V_1/Reg_file1/LTIE_LTIEHI_36_NET
*16 RISC_V_1/Reg_file1/LTIE_LTIEHI_35_NET
*17 RISC_V_1/Reg_file1/LTIE_LTIEHI_34_NET
*18 RISC_V_1/Reg_file1/LTIE_LTIEHI_33_NET
```

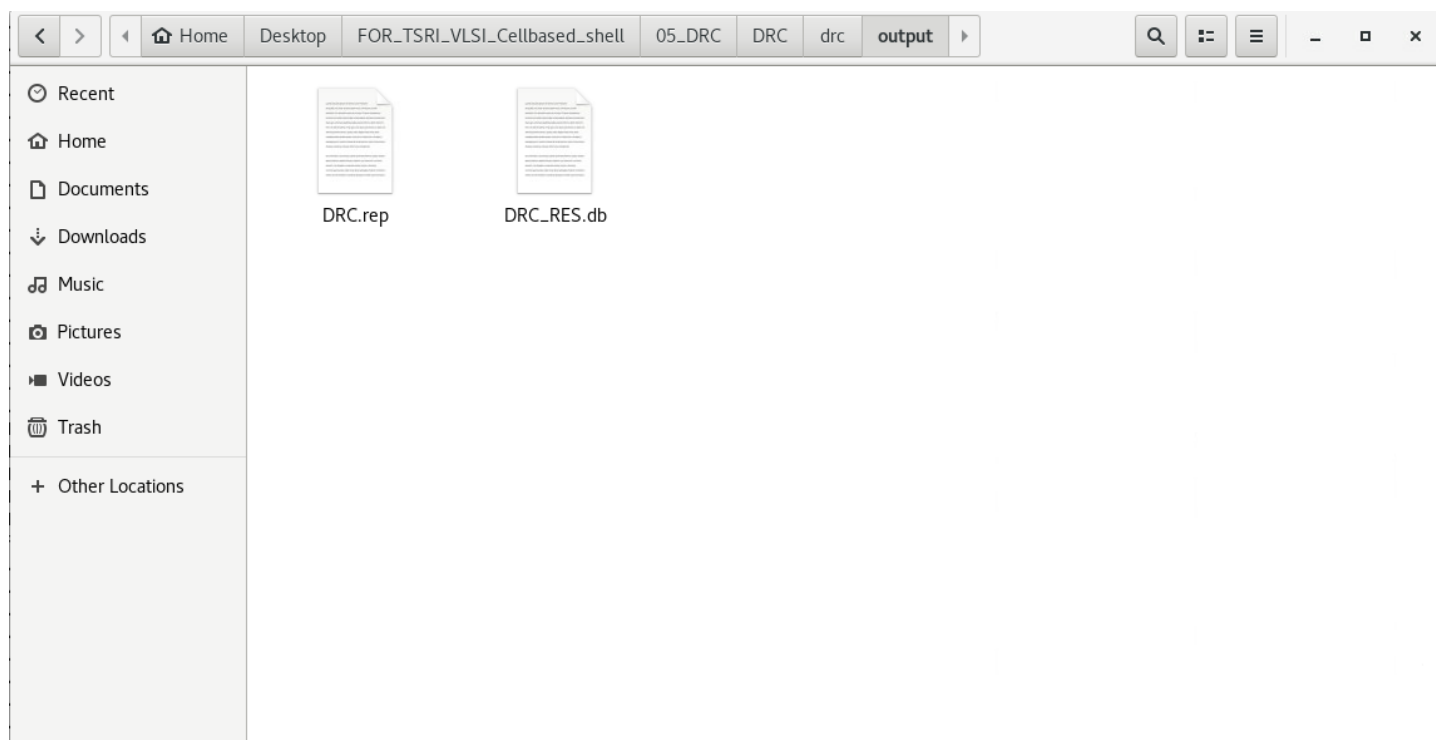
The status bar at the bottom indicates "Plain Text", "Tab Width: 8", "Ln 1, Col 1", and "INS".

31. APR 全部執行結束後 Result 資料夾中的檔案(截資料夾內的 icon)：



32. (06_DRC)DRC_result 資料夾截圖(icon)





33. (07_LVS) lvs.rep 內兩個笑臉區塊截圖

Open

lvs.rep

Save

~/Desktop/FOR_TSRI_VLSI_Cellbased_shell/06_LVS

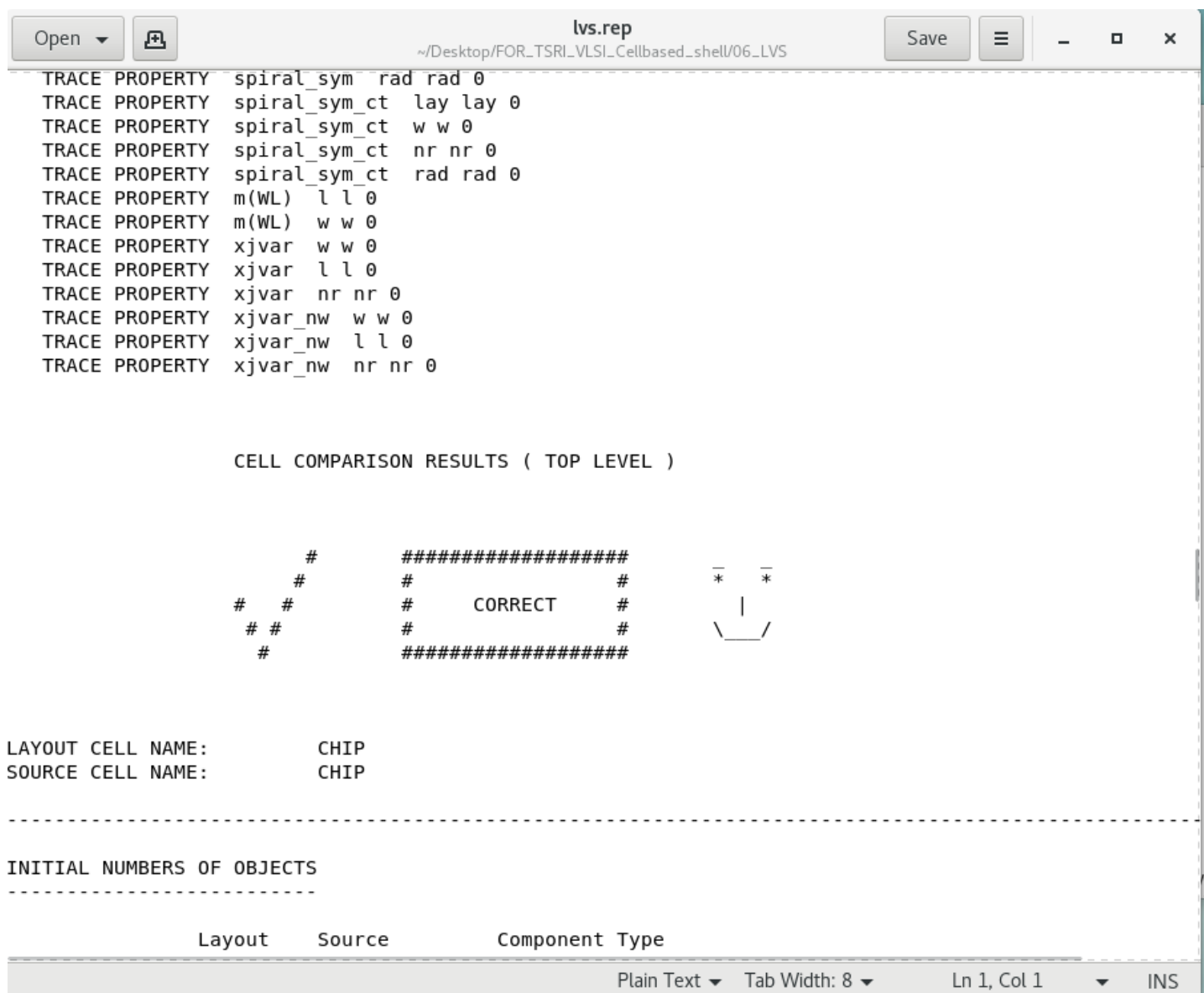
```
#####
##                                ##
##      C A L I B R E   S Y S T E M      ##
##                                ##
##      L V S   R E P O R T      ##
##                                ##
#####

REPORT FILE NAME:      lvs.rep
LAYOUT NAME:          layout.spi ('CHIP')
SOURCE NAME:          source.spi ('CHIP')
RULE FILE:            TN90GUTM_LVS_CB.rule
HCELL FILE:           (-automatch)
CREATION TIME:        Tue Dec 30 17:11:07 2025
CURRENT DIRECTORY:    /cst1/ca004/ca004045/Desktop/FOR_TSRI_VLSI_Cellbased_shell/06_LVS
USER NAME:            ca004045
CALIBRE VERSION:      v2025.2_14.11      Tue Apr 8 13:28:53 PDT 2025

                                OVERALL COMPARISON RESULTS

#                                #####
#                                #
#                                #      *      *
#                                #      |
#                                #      \____/
#                                #
#                                #####
```

Loading file "/cst1/ca004/ca004045/Desktop/FOR_TSRI_VLSI_Cell... Plain Text Tab Width: 8 Ln 1, Col 1 INS



34.(08_LPE_Postsim) 08_LPE_Postsim 資料夾下的 CHIP_POST.sdf
以及 pararistics_command.log 截圖

Open

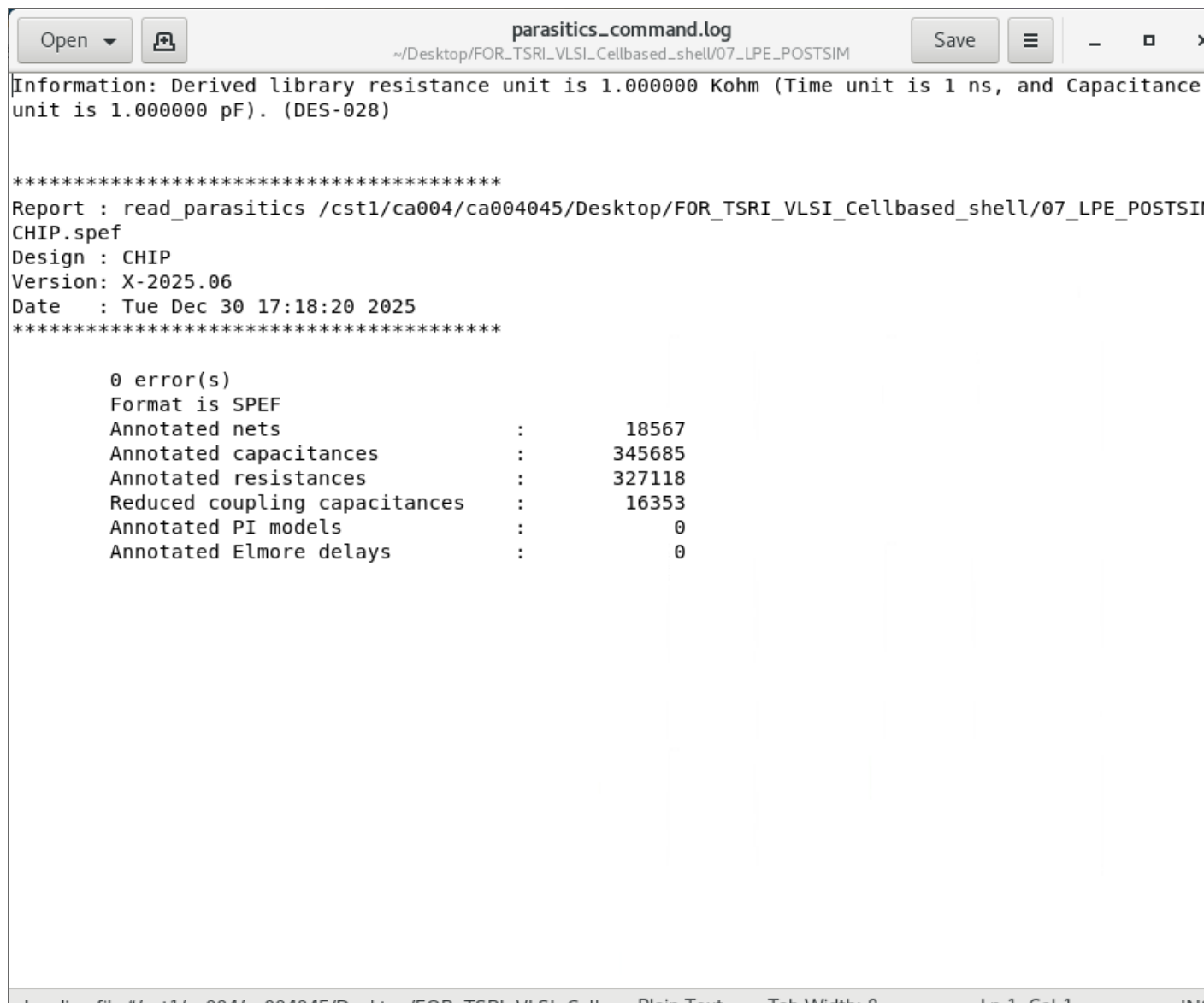
CHIP_POST.sdf

Save

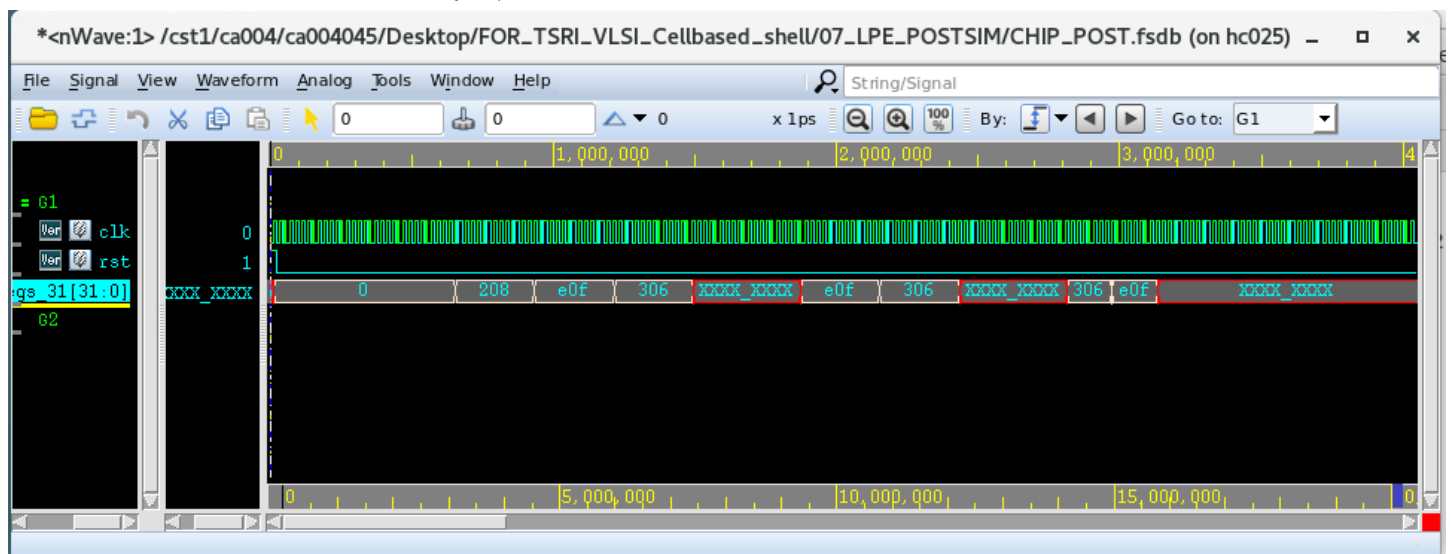
~/Desktop/FOR_TSRI_VLSI_Cellbased_shell/07_LPE_POSTSIM

```
(DELAYFILE
(SDFVERSION "OVI 2.1")
(DESIGN "CHIP")
(DATE "Tue Dec 30 17:18:21 2025")
(VENDOR "slow tpzn90gv3bc")
(PROGRAM "Synopsys PrimeTime")
(VERSION "X-2025.06")
(DIVIDER /)
// OPERATING CONDITION "nom_pvt::nom_pvt"
(VOLTAGE 5.000::5.000)
(PROCESS "1.000::1.000")
(TEMPERATURE 25.000::25.000)
(TIMESCALE 1ns)
(CELL
  (CELLTYPE "CHIP")
  (INSTANCE)
  (DELAY
    (ABSOLUTE
      (INTERCONNECT RISC_V_1/Reg_file1/regs_reg_31_7/Q u_regs_31_7/I (0.000::0.000))
      (INTERCONNECT RISC_V_1/U559/Y RISC_V_1/intadd_0_U19/A (0.000::0.000))
      (INTERCONNECT RISC_V_1/rs2_value_r_reg_29/Q RISC_V_1/U110/B0 (0.000::0.000))
      (INTERCONNECT RISC_V_1/rs1_value_r_reg_26/Q RISC_V_1/U128/A (0.000::0.000))
      (INTERCONNECT RISC_V_1/U535/Y RISC_V_1/U537/A1 (0.000::0.000))
      (INTERCONNECT RISC_V_1/U646/Y RISC_V_1/pc_r_reg_26/D (0.000::0.000))
      (INTERCONNECT RISC_V_1/rom/U155/Y RISC_V_1/U179/AN (0.000::0.000))
      (INTERCONNECT RISC_V_1/U188/Y RISC_V_1/U190/A (0.000::0.000))
      (INTERCONNECT RISC_V_1/U402/Y RISC_V_1/U407/A1 (0.000::0.000))
      (INTERCONNECT RISC_V_1/U56/Y RISC_V_1/U74/B (0.000::0.000))
      (INTERCONNECT RISC_V_1/LUS_1/U235/Y RISC_V_1/U91/B (0.000::0.000))
      (INTERCONNECT RISC_V_1/Reg_file1/U1544/Y RISC_V_1/Reg_file1/U1615/B0 (0.003::0.003)
(0.003::0.003))
      (INTERCONNECT RISC_V_1/Reg_file1/regs_reg_3_10/Q RISC_V_1/Reg_file1/U1627/A1N (0.000::0.000))
      (INTERCONNECT RISC_V_1/Reg_file1/U2391/Y RISC_V_1/Reg_file1/regs_reg_23_9/D (0.000::0.000))
      (INTERCONNECT RISC_V_1/Reg_file1/U876/Y RISC_V_1/Reg_file1/U1647/B1 (0.001::0.001)
(0.000::0.000))
      (INTERCONNECT RISC_V_1/Reg_file1/regs_reg_6_24/Q RISC_V_1/Reg_file1/U1658/B1 (0.000::0.000))
      (INTERCONNECT RISC_V_1/Reg_file1/FE_0FC38_n55/Y RISC_V_1/Reg_file1/U1668/A0 (0.002::0.002)
(0.000::0.000))
    )
  )
)
```

Plain Text Tab Width: 8 Ln 1, Col 1 INS



35. Post-Sim 波形截圖



36. 心得報告

最後因為用自己在計算機系統設計的 RISC-V，中途遇到很多上課沒遇到的事情，一些接腳就要自己接了，但是因為上課做了一遍流程了，所以在自己做的時候除了改檔案的比較難之外，在 DRC 之後就很順利了，沒什麼遇到困難。